

Mitigating Urban Heat Risk in Chennai & Co-Benefits of Green Coverage

Application to Chennai, India

Liufei Chen Vashishth Doshi Farrukh Masood Yuhang Zhou
Carnegie Mellon University · The World Bank Group

May 2026

Contents

1	Introduction	5
1.1	Background	5
1.2	Executive Summary	5
1.3	Case Study Context: Chennai	5
1.4	Project Objectives	6
1.5	Project Deliverables	6
2	Analytical Framework	7
2.1	Framework Overview	7
2.2	Heat Component: SUHII	7
2.3	Social Vulnerability	8
2.4	Exposure	8
2.5	Risk Score Integration	9
2.6	The 3-30-300 Framework for Urban Heat Risk Mitigation	9
2.7	Priority Zones	9
3	Data Description	10
3.1	SUHII Data	10
3.2	Vulnerability & Exposure Data	10
3.3	3-30-300 Data	10
4	Methodology	12
4.1	Spatial Unit Definition	12
4.2	SUHII	12
4.2.1	Rural Baseline Definition	12
4.2.2	SUHII Calculation	12
4.2.3	Aggregation	13
4.2.4	Coastal Considerations	13
4.3	Vulnerability & Exposure	13
4.4	Risk Score	15
4.5	3-30-300	15
4.5.1	Indicator 1: Three Trees Visible	15
4.5.2	Indicator 2: 30% Canopy Cover	16
4.5.3	Indicator 3: 300m Park Access	16
4.6	Priority Zones	16
4.6.1	Integration: Combined Green Index	16
4.6.2	Priority Zones: Overlaying Heat and Green Deficits	17
5	Results	17
5.1	SUHII	17
5.2	Vulnerability and Exposure Patterns	19
5.2.1	Spatial Distribution of Social Vulnerability & Population Exposure	19
5.2.2	Integrated Analysis of Vulnerability and Exposure	19
5.3	Risk Score	21
5.4	3-30-300	23
5.4.1	Three Trees Visible	23

5.4.2	30% Canopy Cover	23
5.4.3	300 m Park Access	24
5.4.4	Combined Greenness Index (3-30-300)	24
5.5	Priority Zones	24
6	Discussion	25
6.1	Key Findings	25
6.2	Planning Implications	26
6.2.1	From Spatial Prioritisation to Implementation	26
6.2.2	The Co-Benefit Value Proposition	26
6.2.3	Participatory Governance	27
6.3	Methodological Contributions and Transferability	27
6.3.1	Halo-Based SUHII Estimation	27
6.3.2	Open-Data 3-30-300 Assessment	27
7	User Research — Expert Consultation Summary	27
7.1	Research Process	28
7.2	Thematic Clustering Matrix	28
7.3	Thematic Synthesis	28
7.3.1	Data, Tools & Methodological Robustness (Themes 1 & 2)	28
7.3.2	Implementation Barriers & Governance (Themes 3 & 4)	28
7.3.3	Co-Benefit Framing & the ROI Imperative (Theme 5)	30
7.3.4	Local Context & Transferability (Themes 6 & 7)	30
7.4	Validated Assumptions & Unresolved Tensions	31
7.5	Practitioner Voices	31
7.6	Implications for the Co-Benefits Analysis	31
8	Co-Benefits Analysis	32
8.1	Introduction	32
8.2	Branch A: Temperature Reduction	32
8.2.1	How Trees Reduce Temperature	32
8.2.2	Research Findings: Canopy and Temperature	32
8.2.3	Research Findings: Temperature and Health	32
8.2.4	Chennai Context	33
8.3	Branch B: Rainfall Interception	33
8.3.1	How Trees Reduce Flooding	33
8.3.2	Research Findings: How Much Rainfall Do Trees Intercept?	33
8.3.3	Research Findings: How Much Does This Reduce Runoff?	33
8.3.4	Chennai Flood Context	34
8.4	Branch C: Air Filtering	34
8.4.1	How Trees Clean Air	34
8.4.2	Research Findings: How Much Pollution Do Trees Remove?	34
8.4.3	Research Findings: Health Impact of Air Pollution Removal	34
8.4.4	Chennai Context	34
8.5	Branch D: Additional Co-Benefits	35
8.5.1	Public Health and Mental Wellbeing	35
8.5.2	Biodiversity and Ecology	35
8.5.3	Economic Benefits and Employment	35

8.6	Connecting the Four Branches	35
8.6.1	Why Co-Benefits Must Be Included in Appraisal	35
8.6.2	Recommended Frameworks for Chennai	36
8.7	Summary of Key Findings	36
8.8	i-Tree Planting Tool: Quantifying Co-Benefits for Chennai	36
8.8.1	What is the i-Tree Planting Tool?	36
8.8.2	Configuration for Chennai	37
8.8.3	Number of Trees and Group Structure	37
8.8.4	Results and Interpretation	38
9	Limitations and Future Work	38
9.1	Data Limitations	38
9.2	Methodological Limits	39
9.3	Future Work	39
	Acknowledgements	41
	References	42

1 Introduction

1.1 Background

Urban heat is a growing challenge for cities worldwide, with built-up areas tending to record higher land surface temperatures than the rural areas around them. The gap is widening as cities expand and global temperatures rise.

The impacts go beyond temperature. Extreme heat increases heat-related mortality and strains public health systems. It also raises building cooling loads, which in turn drives up energy demand and emissions. Streets, markets, and transit stops become harder to use during the hottest hours.

These impacts are not distributed evenly. Following the IPCC risk paradigm (IPCC, 2014), heat risk reflects the interaction of hazard, exposure, and vulnerability. Low-income neighborhoods often face the highest temperatures with the least capacity to adapt, and substandard housing, limited green cover, and weak infrastructure compound the problem. Without a structured assessment, cooling investments tend to land where they are convenient rather than where they are needed.

This motivates a replicable, spatially explicit framework that integrates hazard, exposure, and vulnerability into one composite measure. It should also link the resulting risk surface to actionable green-infrastructure benchmarks such as the 3-30-300 rule.

1.2 Executive Summary

Chennai faces a compounding urban heat crisis. Mean Land Surface Temperature rose by 4.2°C between 2000 and 2020, annual heatwave days are projected to double by 2050, and vegetation has declined by 22% (TNGCC and ICLEI South Asia, 2026).

This report develops a reproducible urban heat risk analysis for Chennai. It combines Surface Urban Heat Island Intensity, Social Vulnerability, and Population Exposure (Doan et al., 2023; IPCC, 2014). The analysis found a city-wide heat island intensity of 5.02°C using a halo-based rural baseline (Raj and Yun, 2024). Social vulnerability concentrates in the north while population exposure peaks in the historic core (Arunachalam et al., 2023). A 2 km² hexagonal grid revealed localized risk pockets that ward-level analysis misses.

The 3-30-300 assessment found severe green coverage deficits across Chennai (Konijnendijk, 2021). None of the 132 hexagons (think of them as neighbourhoods to motivate the granular scale) meet the 3-30-300 green coverage standard. Overlaying heat risk with greenness deficits identified 10 priority intervention zones affecting over an estimated 866,000 residents. User research with local stakeholders grounded these findings in on-the-ground planning realities.

The co-benefits analysis shows that tree canopy expansion reduces temperatures (Ziter et al., 2025), intercepts rainfall (Kuehler et al., 2017), and improves air quality (Nowak et al., 2014). This framework provides policymakers with an evidence-based, spatially explicit tool for targeting green coverage investment where it delivers the greatest public health and climate adaptation outcomes.

1.3 Case Study Context: Chennai

Chennai is the capital of Tamil Nadu and a major coastal metropolis on the Bay of Bengal. The Chennai Metropolitan Area combines a dense historic core, large industrial and residential belts in the north, and rapidly expanding peri-urban zones, with ward-level conditions varying widely in housing quality, infrastructure access, and population density.

Recent climate trends make the case urgent. Mean Land Surface Temperature rose from 29.4°C in 2000 to 33.6°C in 2020, a 4.2°C increase in two decades (TNGCC and ICLEI South Asia, 2026), and annual heatwave days are projected to rise from 42 to 81 by 2050. Vegetation has declined by roughly 22% over the same period, and per capita green cover (8.75 sq.m) sits below the WHO standard of 9.5 sq.m. Around 25% of wards already register very high heat risk, and these wards hold about 30% of the population on just 12% of the city’s area.

Chennai is selected for two specific reasons. First, it is rapidly urbanizing: built-up land in the periphery surged from 11.9% to 44.16% in recent decades (Muthiah et al., 2022). Conventional peri-urban areas can no longer be treated as thermally neutral, so the rural baseline for SUHII must be redefined. Second, the city has strong coastal influence, which can bias urban–rural temperature comparisons if it is not addressed directly.

Neither factor was handled by the previous group whose framework this project builds on, and together they form the novel methodological contribution of this work. Chennai also has reliable ward-level Census data and broad remote-sensing coverage, which supports a replicable analysis for other Indian and South Asian cities.

1.4 Project Objectives

This project builds on the tools and methods developed by the previous group to assess their applicability to Chennai. Specific objectives are: (1) apply and adapt the existing framework (SUHII, vulnerability, and 3-30-300) to Chennai; (2) extend the framework by incorporating population exposure alongside vulnerability to better capture population-level heat risk, while also refining the SUHII component to account for urbanization and coastal effects; and (3) identify priority intervention zones. (4) Quantify the co-benefits of urban tree canopy expansion across temperature reduction, rainfall interception, air filtering, and additional social, ecological, and economic outcomes, so that interventions can be appraised on their full value rather than cooling alone. (5) Ground the framework in user research with local stakeholders and end users to ensure that the priority zones, indicators, and outputs reflect on-the-ground planning needs in Chennai.

1.5 Project Deliverables

All project outputs are archived in the public GitHub repository at https://github.com/Michelle0430/UHI_Chennai. The deliverables are as follows:

- **Reports**

- Integrated Final Report (this document)
- User Research Knowledge Repository — https://github.com/Michelle0430/UHI_Chennai/blob/28d6db584acc1a82f81475aef53113708562aea2/Reports/User%20Research%20Knowledge%20Repository.pdf
- Co-Benefits Analysis White Paper — https://github.com/Michelle0430/UHI_Chennai/blob/28d6db584acc1a82f81475aef53113708562aea2/Reports/Co-Benefits%20Analysis%20White%20Paper.pdf
- Chennai i-Tree Report — https://github.com/Michelle0430/UHI_Chennai/blob/28d6db584acc1a82f81475aef53113708562aea2/Reports/Chennai_iTree_Report.pdf

- **StoryMap & Dashboard**

- Interactive StoryMap — <https://arcg.is/141SLv0>
- Multi-layer Dashboard — <https://www.arcgis.com/apps/dashboards/930d1ea65a6b40ad94fdd68785ddef17>

- **Presentation**

- Recording — <https://cmu.zoom.us/rec/share/vIj3J2AynjAtC-NQEraBs8119P5Kic6jbz-9tPZbLZKelvwTnrQ3Nym3ZgVMYzdc.ZKcs9tepGwGxyWrt?from=hub> (Passcode: B.#r0v6?)

2 Analytical Framework

2.1 Framework Overview

This project adopts an integrated risk assessment and spatial intervention framework, structured according to the classic risk paradigm proposed by the IPCC (IPCC, 2014, 2022). The framework conceptualizes Urban Heat Risk as the intersection of three fundamental components: **Hazard**, **Vulnerability**, and **Exposure**.

The analytical workflow is divided into 4 sequential stages:

1. **Hazard Identification:** Utilizing remote sensing data to derive the Surface Urban Heat Island Intensity (SUHII), which quantifies the physical intensity of thermal stress across the city.
2. **Vulnerability and Exposure:** Assessing Social Vulnerability to understand the inherent sensitivity and adaptive capacity of the population, and Exposure to measure the density of people at risk.
3. **Risk Integration:** Aggregating these 3 dimensions through a multiplicative approach to identify areas facing the highest composite threat.
4. **Priority Action Zoning:** Overlapping the final risk scores with the 3-30-300 green coverage deficit analysis to pinpoint high-priority zones for urban cooling interventions.

2.2 Heat Component: SUHII

Surface Urban Heat Island Intensity (SUHII) is used in this study as the primary indicator of urban heat exposure. SUHII is defined as the difference in mean land surface temperature (LST) between urban areas and a corresponding rural reference area, capturing the extent to which urbanization elevates local thermal conditions (Raj and Yun, 2024).

$$\text{SUHII} = \mu(\text{LST}_{\text{urban}}) - \mu(\text{LST}_{\text{rural}}) \quad (1)$$

where $\text{LST}_{\text{urban}}$ and $\text{LST}_{\text{rural}}$ represent the mean land surface temperatures of urban and rural areas, respectively. This formulation provides a relative measure of urban heat, enabling analysis of spatial variation in heat intensity across different areas, consistent with prior studies on SUHII spatial patterns (Zhou et al., 2014).

2.3 Social Vulnerability

Social vulnerability refers to the socio-economic and institutional factors that influence an individual’s or community’s ability to prepare for, respond to, and recover from extreme heat events (Cutter et al., 2003). In the context of Chennai, vulnerability is driven by a combination of marginalized social status and substandard living conditions. This study utilizes a Principal Component Analysis (PCA) based approach to synthesize 8 variables into a single Social Vulnerability Index (SoVI). These variables represent 2 critical dimensions: Socio-economic Status and Infrastructure Quality (see Table 1).

Table 1: Selected Variables for Social Vulnerability Assessment

Category	Variable Description	Rationale
Socio-economic Status	Percentage of SC/ST Population	Represents historically marginalized groups often lacking social safety nets.
	Illiteracy Rate	Hinders the ability to access and interpret heat-health warning communications.
	Households with No Assets	Indicates a lack of financial capital to afford cooling devices (e.g., AC or fans).
Infrastructure Quality	Percentage of Bad Roofs (Anchor)	Non-permanent materials increase indoor heat exposure due to poor insulation.
	Percentage of Insufficient Drainage Facilities	Correlates with environmental degradation and lower community resilience.
	Percentage of No Clean Water	Increases the risk of dehydration and heat-related illnesses during heatwaves.
	Percentage of Bad Walls	Poor structural materials offer little thermal buffer against high ambient temperatures.
	Percentage of Bad Floors	Typically associated with informal settlements and lower overall housing quality.

While this project relies on granular housing condition data to measure infrastructure quality, we acknowledge that not all these variables are recorded in all other national census records. To ensure the generalizability of this PCA-based framework, future applications can substitute these specific structural variables with widely available proxies. Variables such as the percentage of mobile homes, severe housing cost burden, overcrowding, and lack of AC/HVAC access can effectively proxy for poor infrastructure quality. Additionally, universal socio-economic metrics like median household income, unemployment rates, and education level can seamlessly integrate into the PCA to synthesize a comparable vulnerability score tailored to local data availability.

2.4 Exposure

Exposure is defined as the presence of people, livelihoods, or assets in places that could be adversely affected by heat hazards (IPCC, 2014). This project uses population density (persons per sq. km) at the Ward level as the primary proxy for exposure.

High population density amplifies the potential public health impact of an UHI hazard. Even in areas with lower vulnerability, extreme density can lead to high total risk due to the sheer number of individuals exposed to the hazard (Doan et al., 2023).

2.5 Risk Score Integration

The Urban Heat Risk Score is calculated by integrating heat exposure (SUHII) with the Population Risk component into a single composite metric, where Population Risk reflects the combined effect of exposure and social vulnerability.

$$\text{Urban Heat Risk} = \text{SUHII} \times \text{Social Vulnerability} \times \text{Exposure} \times 100 \quad (2)$$

This multiplicative formulation ensures that areas with simultaneously high heat exposure and high population risk are prioritized in the assessment. If any component is low, the overall risk is moderated, reflecting the interaction between environmental stress and social vulnerability. This formulation follows widely used climate risk frameworks in which overall risk is understood as the interaction between hazard, exposure, and vulnerability (Doan et al., 2023).

2.6 The 3-30-300 Framework for Urban Heat Risk Mitigation

Urban heat vulnerability results from the intersection of physical exposure and inadequate cooling infrastructure. In rapidly urbanizing cities like Chennai, where the mean NDVI of 0.110 signals severe vegetation deficit, green coverage gaps compound heat-related health risks. We adopt the 3-30-300 framework (Konijnendijk, 2021) to systematically assess where green coverage deficits coincide with heat vulnerability, enabling targeted intervention.

The 3-30-300 rule establishes three evidence-based standards for healthy, livable neighborhoods. First, every resident should see at least three trees from their home - visual contact with nature correlates with mental wellbeing and neighborhood satisfaction (Konijnendijk, 2023). Second, neighborhoods should maintain at least 30% canopy cover - the threshold where measurable cooling effects, stormwater absorption, and air quality improvements emerge (Bowler et al., 2010). Third, everyone should live within 300 meters of quality green space - walkable access to parks supports physical activity, social gathering, and recreational needs.

These indicators work as an integrated system. Tree visibility addresses individual perception of nature; canopy cover provides neighborhood-scale ecosystem services; park access ensures equitable distribution of public green space. Together, they capture both presence of vegetation and accessibility to greenery across spatial scales - a multi-benefit approach increasingly recognized in urban greening policy (Alves et al., 2019).

2.7 Priority Zones

For Chennai, this framework reveals where green coverage is most urgently needed. We calculate a combined greenness index averaging the three indicators, then overlay it with heat risk scores (SUHII methodology from Zhou et al. (2019); social vulnerability from Cutter et al. (2003)) to identify compounded vulnerability. Areas scoring in the bottom 50% for greenness while ranking in the top 25% for heat risk become priority intervention zones - neighbourhoods experiencing both severe vegetation deficit and elevated heat exposure affecting vulnerable populations.

This integrated approach moves beyond single-metric assessments. The 3-30-300 framework captures nuance while the heat overlay ensures interventions address climate adaptation alongside general livability (Ruangpan et al., 2020). The result: an evidence-based spatial framework identifying where green coverage investment delivers maximum co-benefits for heat mitigation and community health.

3 Data Description

3.1 SUHII Data

The estimation of SUHII in this study relies on multiple geospatial datasets that capture surface temperature, vegetation conditions, land cover characteristics, and topographic variation within the study area. As summarized in Table 2, these datasets provide the foundation for SUHII estimation.

Table 2: Summary of Geospatial Datasets Used for SUHII Estimation

Data Type	Dataset	Purpose
Administrative Boundary	Chennai administrative boundary (GADM)	Defines the spatial extent of the study
Land Surface Temperature	Landsat 8 and 9 Collection 2 Level 2	Measures surface heat conditions for SUHII estimation
Vegetation Index (NDVI)	Sentinel-2 Harmonized	Identifies vegetation cover
Land Use / Land Cover	Dynamic World V1	Distinguishes urban and non-urban areas
Elevation	SRTM DEM	Accounts for topographic variation

NDVI and LULC were derived for a consistent observation date to ensure temporal alignment across datasets, while LST was constructed as a seasonal composite to represent prevailing thermal conditions.

3.2 Vulnerability & Exposure Data

The socio-demographic and housing data for this study were primarily sourced from the **Census of India (2011)** ([Office of the Registrar General & Census Commissioner, India, 2011](#)). This national census dataset was selected to ensure the representativeness of the data and the replicability of the analytical framework across other urban contexts. Data were extracted at the Ward level for the Chennai Metropolitan Area (CMA), encompassing both the **Primary Census Abstract (PCA-series)** and the **Housing Census (H-series)**. These attributes were spatially joined to the Chennai administrative boundary shapefile using unique Ward IDs. The definitions, formulas, and criteria for each variable are detailed in Table 3.

3.3 3-30-300 Data

Table 4 summarizes the geospatial datasets used for the 3-30-300 green coverage assessment.

Coordinate System: All spatial data reprojected to WGS 1984 UTM Zone 44N (EPSG: 32644) for accurate metric distance and area calculations.

Data Management: NULL vs. zero distinction: NULL = no data/outside study area; 0 = measured zero value. Decimal precision: scores stored with 6 decimal places; statistics rounded to 3 for reporting. The green coverage analysis was conducted using ArcGIS Pro and shared as web layers to create a dashboard.

Table 3: Variables for Social Vulnerability and Exposure Assessment

Variable Name	Formula	Definition & Criteria
Illiteracy Rate	Illiterate Pop. / Total Pop. Older than 7	Per census definitions, children below age 7 are excluded from literacy counts.
SC/ST Rate	(SC + ST Pop.) / Total Pop.	Proportion belonging to Scheduled Castes or Scheduled Tribes.
Bad Roof Rate	HH with Bad Roof / Total HH	Roofs of non-permanent (Katcha) materials.
Bad Wall Rate	HH with Bad Wall / Total HH	Walls of grass, thatch, bamboo, mud, unburnt bricks, or wood.
Bad Floor Rate	HH with Bad Floor / Total HH	Floors made of mud, wood, or bamboo.
No Clean Water %	HH with No Safe Water / Total HH	Primary water from untreated taps, uncovered wells, or non-potable sources.
No Drainage %	HH with Insuff. Drainage / Total HH	Open drainage or total lack of drainage facilities.
No Assets %	HH with No Assets / Total HH	No specified census assets (radio, TV, computer, phone, bicycle, or vehicle).
Population Density	Total pop. / Area per km ²	Primary proxy for the scale of human exposure.

Note: Pop. denotes population. HH denotes households.

Table 4: Summary of Geospatial Datasets Used for the 3-30-300 Green Coverage Assessment

Dataset	Source	Specifications	Usage	Limitations
Landsat 9 NDVI	USGS EarthExplorer	Scene: LC09_L2SP_142051...; Date: Apr 24, 2025; 30 m resolution; Cloud <5%	Tree visibility (focal NDVI, 50 m radius, >0.35); Canopy cover (binary mask >0.4)	30 m resolution may miss small green spaces (<900 m ²)
Parks Dataset (629 locations)	OpenCity Foundation & GCC (2024)	Point features; Small (<4,000 m ²), Medium (4,000–8,000), Large (>8,000); WGS 1984 UTM 44N	300 m park access (network service area, graduated scoring)	Quality, maintenance, accessibility barriers not captured
Road Network	OSM via LivingAtlas/ArcGIS	Polyline features	Network distance for 300 m park accessibility	Informal pedestrian paths not represented
Urban Heat Risk Scores	Analysis outcomes as explained in sections 2.2-2.5	Range 0–356.912; Mean 81.612; Q3 =134.458	Overlay for priority zones (low greenness AND high heat risk)	Based on 2011 Census (14-year lag)

4 Methodology

4.1 Spatial Unit Definition

The spatial unit of analysis is defined using a hexagonal tessellation to ensure consistent and comparable spatial representation across the study area (Figure 1a). A 2 km hexagonal grid was applied to the Chennai administrative boundary to establish uniform spatial units for analysis.

Ward-level units in Chennai vary considerably in size and shape, which can lead to inconsistent spatial representation. Using a regular grid helps ensure more consistent spatial comparison.

To preserve spatial consistency, a uniform hexagonal grid was applied without modifying the geometry of individual cells along the study boundary. This ensures that each spatial unit maintains a consistent shape and area, including those located along the boundary. We call these spatial units neighbourhoods for readability. Avoiding cell-level clipping prevents the creation of irregular edge geometries and reduces potential bias in spatial aggregation.

4.2 SUHII

This section describes the methodology used to compute SUHII, including rural baseline definition, pixel-level calculation, spatial aggregation, and coastal considerations.

4.2.1 Rural Baseline Definition

Defining an appropriate rural baseline is critical for estimating SUHII, as it establishes the reference against which urban temperatures are compared. In Chennai, rapid urban expansion has significantly altered peri-urban areas. These areas can no longer be considered thermally neutral or representative of rural conditions. Previous studies in the Chennai Metropolitan Area show a substantial increase in built-up land, along with a clear rise in land surface temperature in peri-urban regions (Muthiah et al., 2022). This indicates that peri-urban areas are already influenced by urban heat and should not be used as a rural reference.

To address this issue, a halo-based approach was adopted to minimize peri-urban influence. A 10 km exclusion zone was defined beyond the urban boundary to avoid peri-urban effects. Rural pixels were then sampled within a 15 km band beyond this zone, corresponding to a 10–25 km range from the urban edge (Figure 1b). This configuration was selected based on sensitivity analysis, which showed minimal variation in the rural baseline across different halo-buffer configurations.

4.2.2 SUHII Calculation

Using the defined rural baseline, SUHII was calculated at the pixel level as the difference between land surface temperature (LST) at each pixel and the mean LST of the rural reference area:

$$\text{SUHII}_{\text{pixel}} = \text{LST}_{\text{pixel}} - \mu(\text{LST}_{\text{rural}}) \quad (3)$$

The calculation was implemented using a previously developed SUHII analysis tool (Sah, 2025). The output is a pixel-level raster representing temperature differences across the study area.

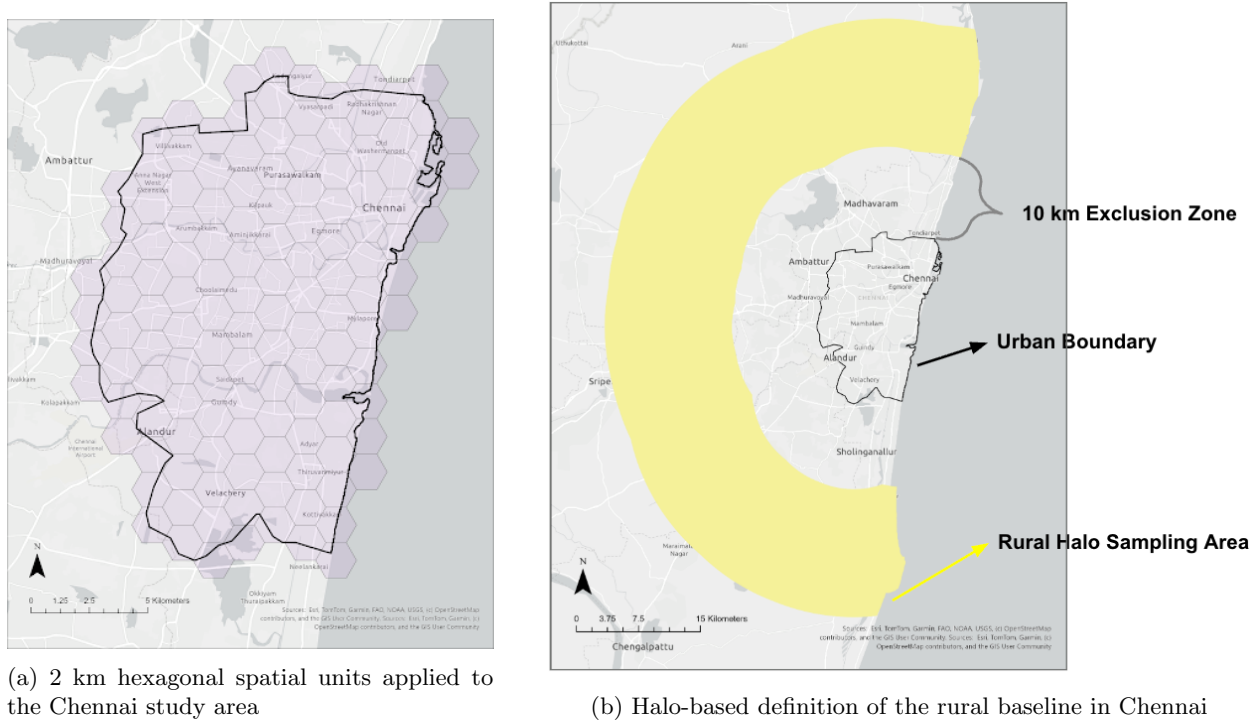


Figure 1: Spatial setup of the SUHII analysis

4.2.3 Aggregation

The pixel-level SUHII raster was aggregated to the predefined spatial units using zonal statistics. For each unit, the mean SUHII value was calculated to represent the average temperature difference. This aggregation reduces local variability and supports consistent spatial comparison across the study area.

4.2.4 Coastal Considerations

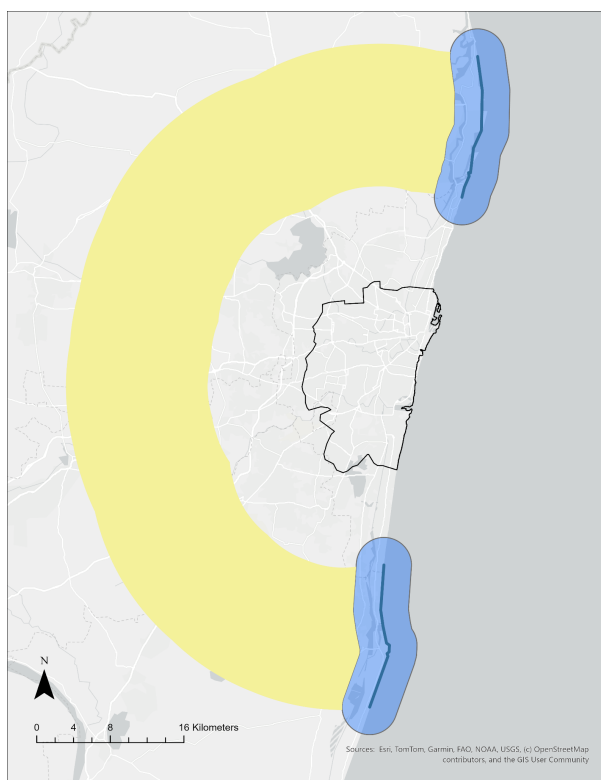
Given the coastal setting of the study area, proximity to the coastline can influence land surface temperature patterns. Prior research indicates that proximity to water bodies can affect temperature conditions, and that urban and rural reference areas should be located at comparable distances from the coastline to avoid bias in UHI estimation (Stewart and Mills, 2021). In this study, the selected rural reference area also includes coastal-adjacent regions, similar to the urban area. As a result, both areas are subject to comparable coastal influence.

To further examine this effect, a sensitivity analysis was conducted by excluding rural pixels within 3 km and 5 km of the coastline (Figure 2). This approach is used to assess whether coastal proximity materially influences the definition of the rural baseline.

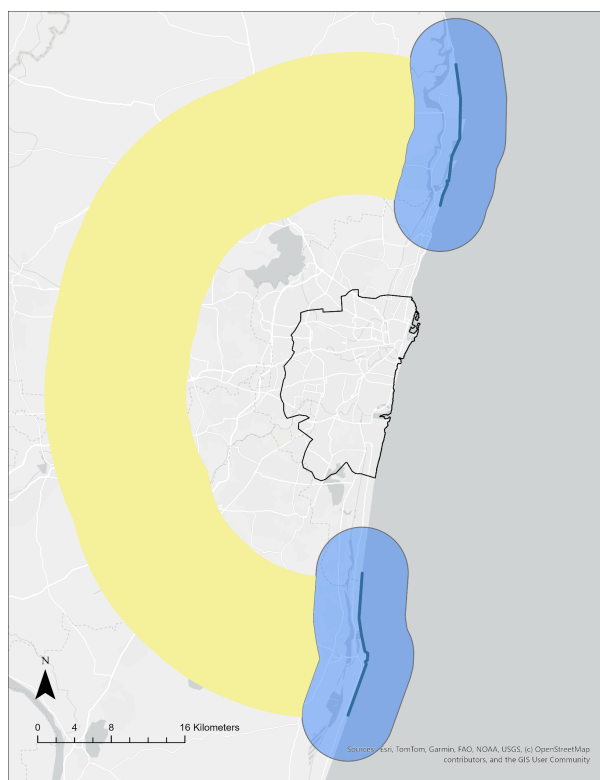
4.3 Vulnerability & Exposure

This project employs a Principal Component Analysis (PCA) based approach to construct the Social Vulnerability Index (SoVI) and standardizes population exposure for risk integration.

Principal Component Analysis: Socio-economic and housing variables often exhibit high multicollinearity. For instance, substandard housing materials are frequently correlated with lower asset



(a) 3 km coastal exclusion zone



(b) 5 km coastal exclusion zone

Figure 2: Rural sampling configuration with coastal exclusion zones

ownership and higher illiteracy. Assigning equal weights to such variables would lead to information redundancy and biased results. PCA addresses this by transforming correlated variables into a set of uncorrelated (orthogonal) principal components. This method retains the maximum variance of the original data while eliminating redundancy, allowing for a more objective assignment of weights based on the data structure (Arunachalam et al., 2023).

Normalization: The raw variables utilized in this study vary significantly in scale and unit (e.g., population density is an absolute value, while housing indicators are percentages). Normalization, particularly Min-Max scaling, is essential to ensure that all indicators are comparable and mapped to a $[0, 1]$ range. This prevents variables with larger numerical scales from disproportionately dominating the final risk score equation (Doan et al., 2023).

Computational Workflow: (i) All raw vulnerability variables were standardized using Z-score transformation. (ii) PCA was performed to extract components with eigenvalues greater than 1.0. Following extraction, the Bad Roof Rate was utilized as a Vulnerability Anchor to calibrate the directionality of the components. Factor loadings were examined to ensure that the signs of the components aligned such that higher values consistently represent higher vulnerability. (iii) The calibrated components were summed to produce the final SoVI for each Ward, which was then normalized for the risk integration stage. (iv) The population exposure index is calculated by normalized population density at the ward level.

4.4 Risk Score

The Urban Heat Risk Score is computed by integrating SUHII, Social Vulnerability Index (SoVI), and Population Exposure. All input layers were aligned to a common hexagonal grid, enabling direct combination of the three components.

The integration is performed using a consistent multiplicative structure across all grid cells. The resulting composite surface represents the spatial distribution of combined heat intensity, vulnerability, and exposure, and serves as the basis for subsequent spatial analysis and prioritization.

4.5 3-30-300

Translating the 3-30-300 conceptual framework into quantitative spatial indicators required confronting a fundamental challenge: satellite imagery cannot count individual trees or distinguish a pocket park from a residential lawn. We developed proxy methods that balance scientific rigor with data availability, prioritizing approaches that could scale to other cities without requiring proprietary datasets or extensive field surveys.

4.5.1 Indicator 1: Three Trees Visible

The ideal implementation would involve a tree census documenting location, species, and canopy dimensions - data Chennai does not yet possess. Instead, we turned to a spatial pattern approach: if three trees are visible from a window, they likely form a cluster of dense vegetation detectable in Landsat’s 30-meter pixels.

We applied Focal Statistics in ArcGIS Pro to calculate the mean NDVI within a 50-meter radius moving window - approximating a typical sightline from a residence. Research demonstrates that NDVI values exceeding 0.35 reliably indicate clustered woody vegetation rather than scattered grass (Weier and Herring, 2000). We rasterized this threshold ($NDVI \geq 0.35 = 1$, otherwise 0) and used Zonal Statistics to calculate the proportion of each hexagon meeting the visibility criterion.

A hexagon scoring 0.80 means 80% of its area contains focal NDVI values suggesting visible tree clusters - our proxy for the “3 trees visible” standard.

This approach sacrifices precision (we estimate visibility, not count trees) but gains scalability: any city with Landsat coverage can replicate this analysis. The 50-meter radius balances Landsat’s resolution with typical residential viewing distances ([Konijnendijk, 2023](#)).

4.5.2 Indicator 2: 30% Canopy Cover

Distinguishing tree canopy from grass or shrubs demands a higher NDVI threshold than the visibility analysis. Literature on urban vegetation mapping consistently identifies NDVI values between 0.4 and 0.7 as indicative of dense, healthy tree canopy - vegetation tall and woody enough to provide cooling and stormwater benefits ([Carlson and Ripley, 1997](#)). Below 0.4, pixels increasingly capture grass, crops, or sparse shrubs; above 0.7 suggests measurement saturation rather than denser vegetation.

We created a binary canopy mask using Raster Calculator: pixels with $NDVI > 0.4$ coded as canopy (1), all others as non-canopy (0). Zonal Statistics computed the mean value per hexagon, yielding canopy percentage. A neighbourhood with mean = 0.23 contains 23% canopy coverage. We normalized scores to the 30% standard: $canopy_30 = canopy_pct/30.0$, capped at 1.0. A perfectly compliant hexagon ($\geq 30\%$ canopy) scores 1.0; a hexagon with 15% canopy scores 0.5.

The fixed threshold (0.4) simplifies implementation but cannot adapt to local vegetation phenology. April imagery minimizes monsoon green-up effects that inflate grass NDVI, focusing analysis on established woody vegetation.

4.5.3 Indicator 3: 300 m Park Access

Measuring park access confronted a critical design choice: Euclidean buffers (circles radiating 300 meters from parks) assume people walk through buildings and across highways. Network distance - walking along actual roads - reflects pedestrian reality. We chose network analysis despite its complexity because accessibility must account for Chennai’s infrastructure barriers: the Cooum River, broad roads without pedestrian crossings, and industrial zones with no through-streets.

Using ArcGIS Network Analyst, we generated service areas (300-meter walking distance) from all 629 park centroids, assuming an average walking speed of 5 km/h along the validated OpenStreetMap road network ([OpenStreetMap contributors, 2024](#)). We performed a Spatial Join between service area polygons and hexagons, calculating the percentage of each hexagon’s area within 300 meters of at least one park.

Binary scoring (access vs. no access) would ignore partial coverage - a hexagon with 5% coverage differs meaningfully from one with 0%. We implemented graduated scoring: $\geq 15\%$ coverage $\rightarrow$ score = 1.0; $\geq 10\% \rightarrow 0.9$; $\geq 5\% \rightarrow 0.75$; $\geq 2\% \rightarrow 0.5$; $< 2\% \rightarrow$ score proportional to percentage.

4.6 Priority Zones

4.6.1 Integration: Combined Green Index

We averaged the three indicators with equal weighting: $greenness_333 = (tree_vis + canopy_30 + park_300)/3$. This equal weighting reflects the 3-30-300 framework’s premise that visual contact, canopy benefits, and park access contribute independently to livability ([Konijnendijk, 2021](#)). A

neighborhood might have excellent park access (high “300” score) but lack street trees (low “3” and “30” scores), or vice versa - the combined index captures this multidimensional deficit.

Scores should range from 0 (fails all three standards) to 1.0 (meets all three). We find that they don’t since no neighbourhood meets all criteria. We calculated the median across all 132 hexagons: 0.030. This threshold - the 50th percentile - becomes our definition of “low greenness.”

4.6.2 Priority Zones: Overlaying Heat and Green Deficits

Green infrastructure deficit pervades Chennai - mean NDVI of 0.110 indicates city-wide scarcity. Addressing the worst half (66 hexagons below 0.030) casts a broad intervention net appropriate to the scale of need. Heat risk, conversely, concentrates in specific neighborhoods where physical exposure (SUHII) compounds social vulnerability (Cutter et al., 2003). Targeting the top quartile (33 hexagons exceeding 134.458) focuses resources where thermal stress intersects vulnerable populations.

We performed sequential selections in ArcGIS: Select by Attributes identified hexagons where greenness_333 < 0.030, then a second query selected hexagons where heat_risk > 134.458. The Intersect operation yielded 10 hexagons - neighborhoods experiencing compounded vulnerability from inadequate cooling infrastructure and elevated heat exposure. These became our priority intervention zones.

5 Results

5.1 SUHII

The spatial distribution of SUHII shows that most urban areas are consistently warmer than the selected rural baseline across the study area (Figure 3). Most spatial units exhibit positive SUHII values, reflecting a clear urban-rural temperature difference. The majority of grid cells fall within the 3–5°C and 5–7°C ranges, with these intervals representing the dominant range of SUHII values across the study area.

The robustness of the rural baseline was assessed across multiple halo configurations (Table 5). Both rural mean temperature and SUHII values vary slightly across configurations, with the magnitude of change decreasing as the sampling range extends outward. For example, the rural mean LST decreases by approximately 0.279°C from the 2–10 km to the 10–10 km configuration, while extending to 10–15 km results in a negligible change of about 0.006°C. A similar pattern is observed for SUHII values, where differences become minimal at larger distances. This diminishing change suggests that the baseline is approaching a stable plateau. Based on this trend, the 10–15 km configuration is considered to provide a reliable and consistent rural reference for SUHII estimation.

Table 5: Sensitivity of SUHII estimates under different rural baseline configurations.

Halo Configuration (Exclusion–Halo Width)	Rural Mean LST (°C)	Δ LST (°C)
2 km–10 km	40.482	—
5 km–10 km	40.440	–0.042
10 km–10 km	40.204	–0.236
10 km–15 km	40.198	–0.006

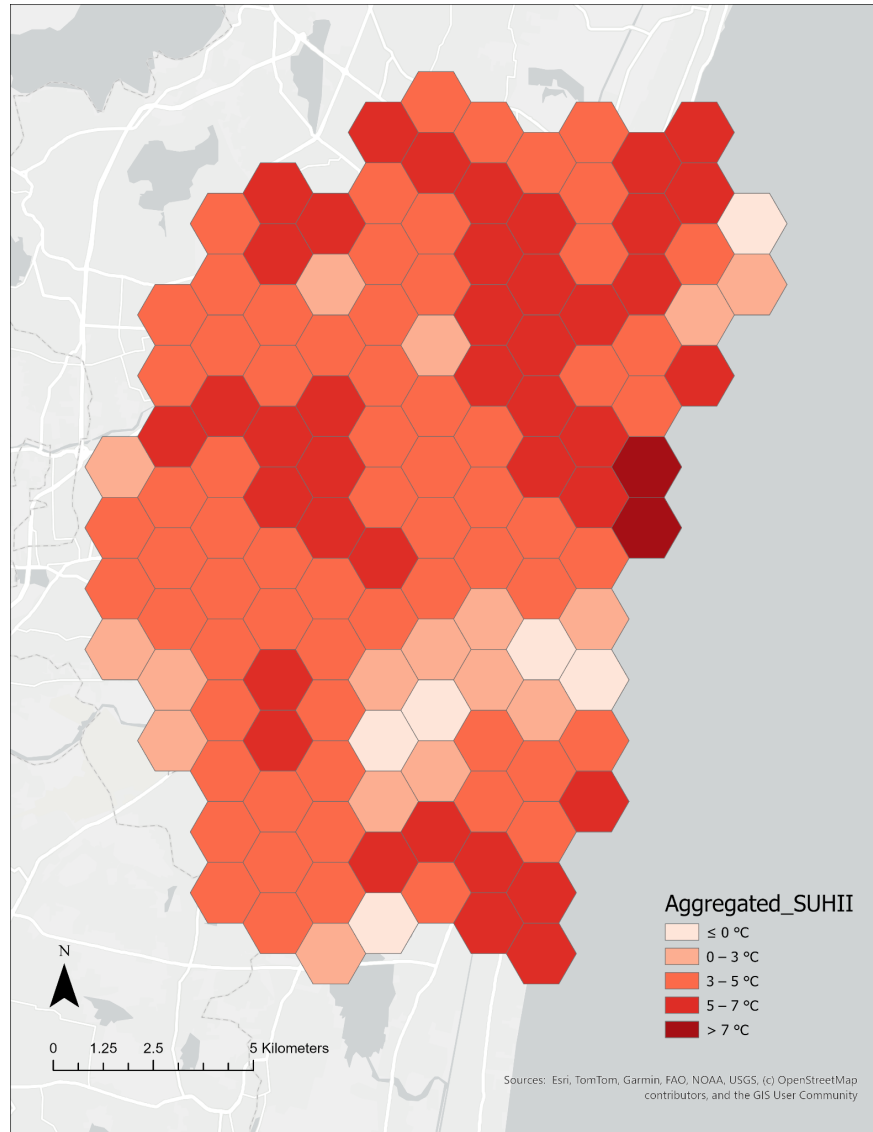


Figure 3: Spatial distribution of aggregated SUHII values

The sensitivity analysis indicates that imposing coastal exclusion constraints produces only limited changes in the rural baseline temperature (Table 6). Relative to the baseline scenario, the mean rural LST decreases from 42.147°C to 42.081°C under the 3 km exclusion and to 42.012°C under the 5 km exclusion, corresponding to changes of 0.066°C and 0.135°C, respectively. Given the small magnitude of these differences, coastal proximity has a limited influence on the rural baseline definition. The selected rural baseline configuration can therefore be considered sufficiently robust for SUHII estimation.

Table 6: Sensitivity of rural baseline temperature to coastal exclusion distances.

Scenario	Rural Mean LST (°C)	Δ LST (°C)
Baseline	42.147	—
3 km	42.081	−0.066
5 km	42.012	−0.135

Overall, the rural baseline is defined within a stable range based on the sensitivity analysis. The coastal analysis suggests that rural areas in close proximity to the coastline do not substantially reduce temperatures, and coastal-adjacent areas are slightly warmer than inland locations. This indicates that rural areas near the coastline do not show a clear reduction in temperature associated with marine influence. This supports the reliability of the SUHII estimates.

5.2 Vulnerability and Exposure Patterns

5.2.1 Spatial Distribution of Social Vulnerability & Population Exposure

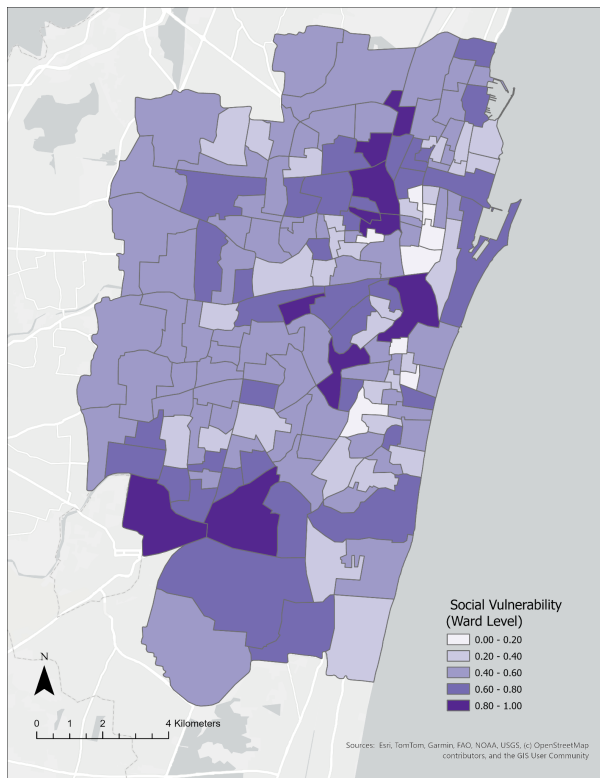
In the following maps, darker shades represent higher numerical scores, indicating either increased social vulnerability or higher population density.

The spatial pattern of SoVI reveals a pronounced North-South divide within Chennai. As illustrated in Figure 4(a), high-vulnerability clusters (represented by dark purple shades) are predominantly concentrated in the northern and northwestern fringes, such as Tiruvottiyur and its surrounding wards. These areas are characterized by a high prevalence of informal settlements, substandard housing materials (e.g., poor roofing), and limited socio-economic assets. Conversely, the southern coastal wards exhibit significantly lower SoVI scores, reflecting better infrastructure and higher adaptive capacity among residents.

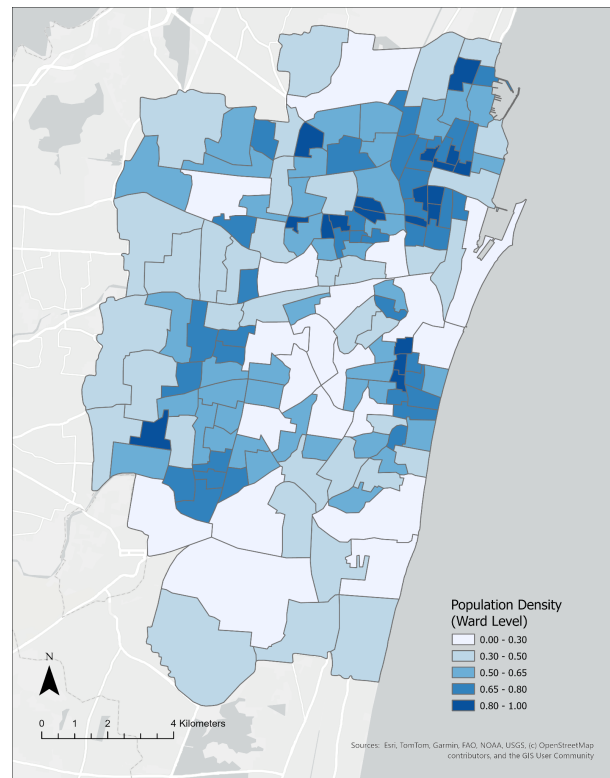
In contrast to the vulnerability patterns, population exposure is driven by urbanization density rather than socio-economic deprivation. As shown in Figure 4(b), the highest exposure levels (represented by dark blue shades) are found in the historic core and central business districts (Old City). These central wards exhibit extreme population density, meaning that even if their inherent social vulnerability is moderate, the sheer volume of individuals residing in these heat-stressed environments creates a substantial public health challenge.

5.2.2 Integrated Analysis of Vulnerability and Exposure

To identify the most critical areas for intervention, we calculated a combined Vulnerability-Exposure Index by multiplying Normalized Social Vulnerability by Normalized Population Exposure. Rather than serving strictly as a heat-specific metric, this integrated layer functions as a broader baseline of urban vulnerability. It universally highlights areas susceptible to multiple intersecting climate and urban stresses, such as extreme heat, pluvial flooding, and air pollution. In this context, the



(a) Social Vulnerability Index (SoVI) at the Ward Level



(b) Population Exposure at the Ward Level

Figure 4: Spatial distribution of Social Vulnerability and Population Exposure

highest scores, which are represented by the darkest brown shades on the maps, signify systemic zones under double threat. These are areas burdened by both massive population concentration and deep socio-infrastructural disadvantages, making them fundamentally less resilient to any hazard.

Spatially, Figure 5(a) illustrates this baseline at the ward level, showing heavy risk concentrations in the northern industrial and residential belts. However, by downscaling this analysis to a high-resolution grid (Figure 5(b)), our framework identifies micro-scale islands of risk that are typically obscured by ward-level averages. This spatial granularity reveals localized pockets of extreme distress even within seemingly resilient wards, providing precise coordinates for targeted climate adaptation strategies.

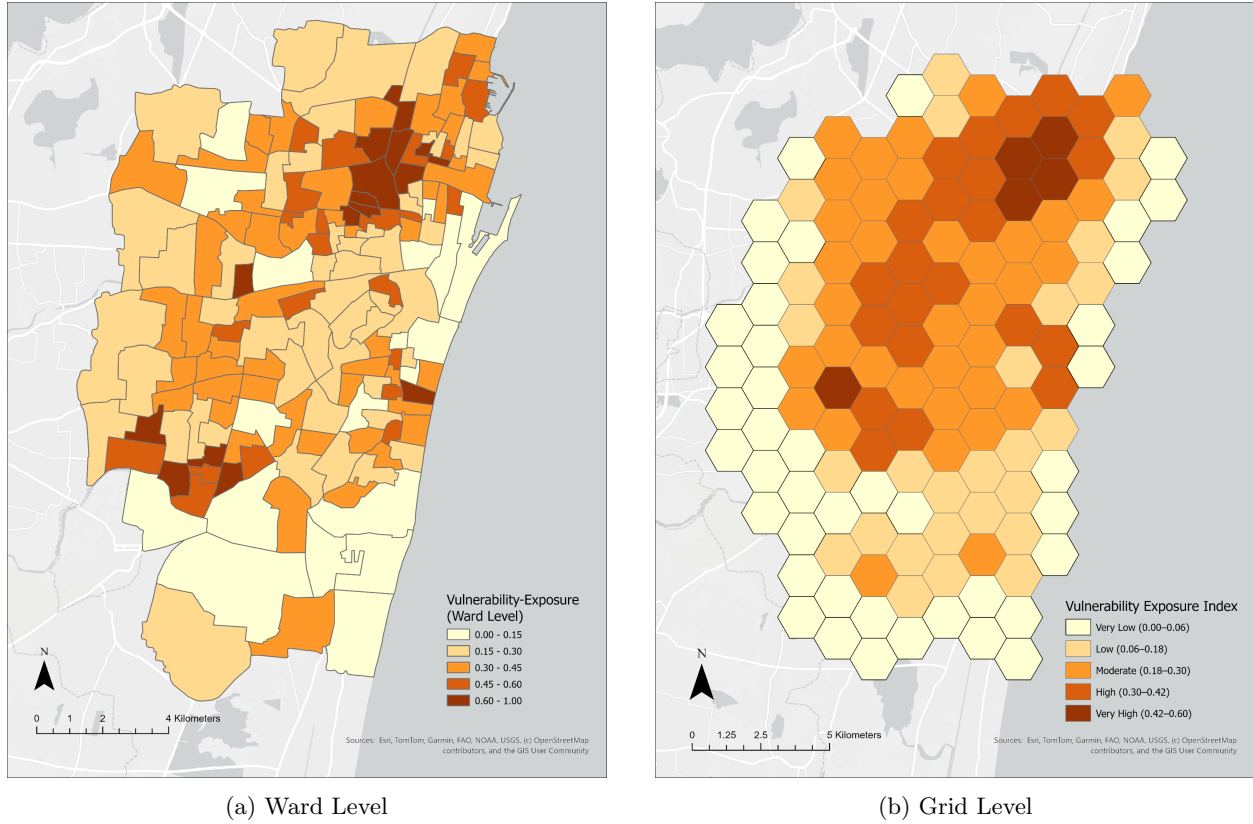


Figure 5: Integration of Social Vulnerability and Population Exposure

5.3 Risk Score

The spatial distribution of the Urban Heat Risk Score shows clear variation across the study area (Figure 6). The majority of spatial units fall within the 24–123 and 123–194 ranges. Higher values (194–356) appear in several localized clusters, while lower values (0–24) are mainly found in peripheral areas. High-risk areas are observed in locations where heat exposure, vulnerability, and population levels are all relatively high.

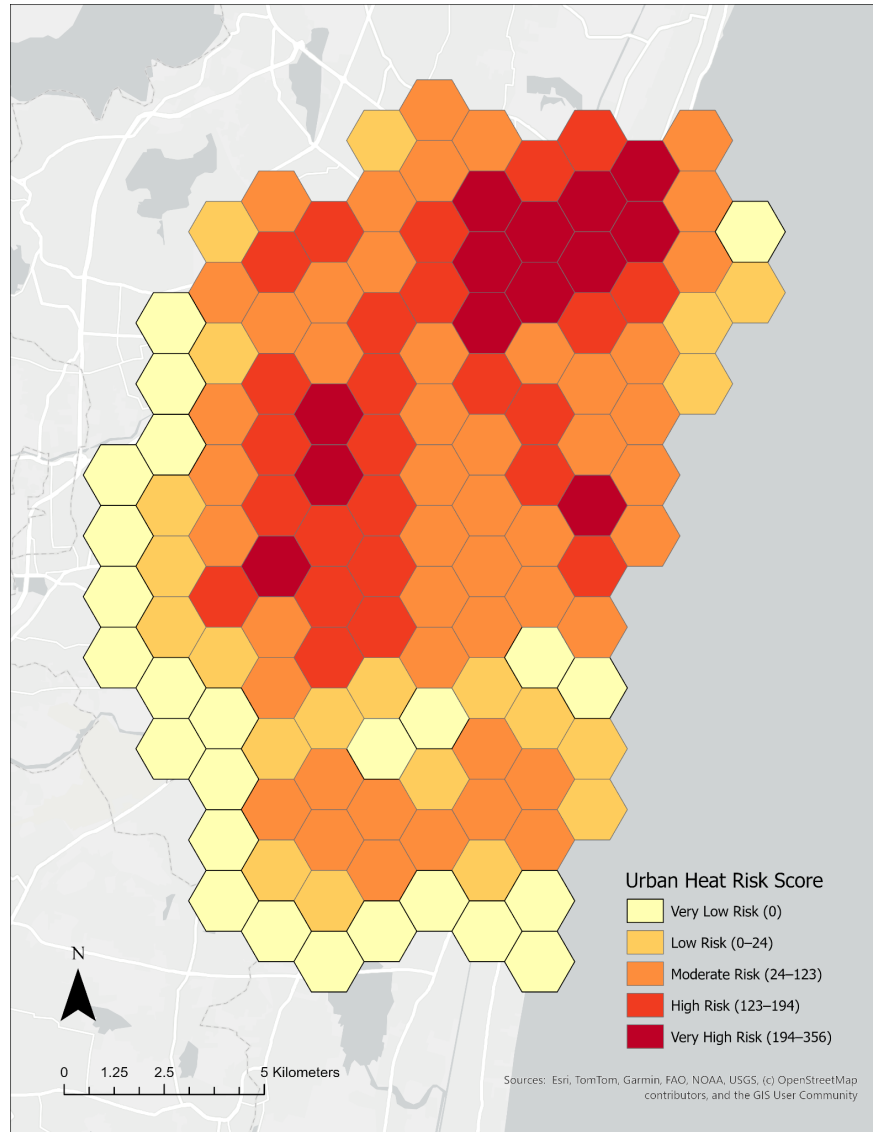


Figure 6: Spatial distribution of the Urban Heat Risk Score across the study area

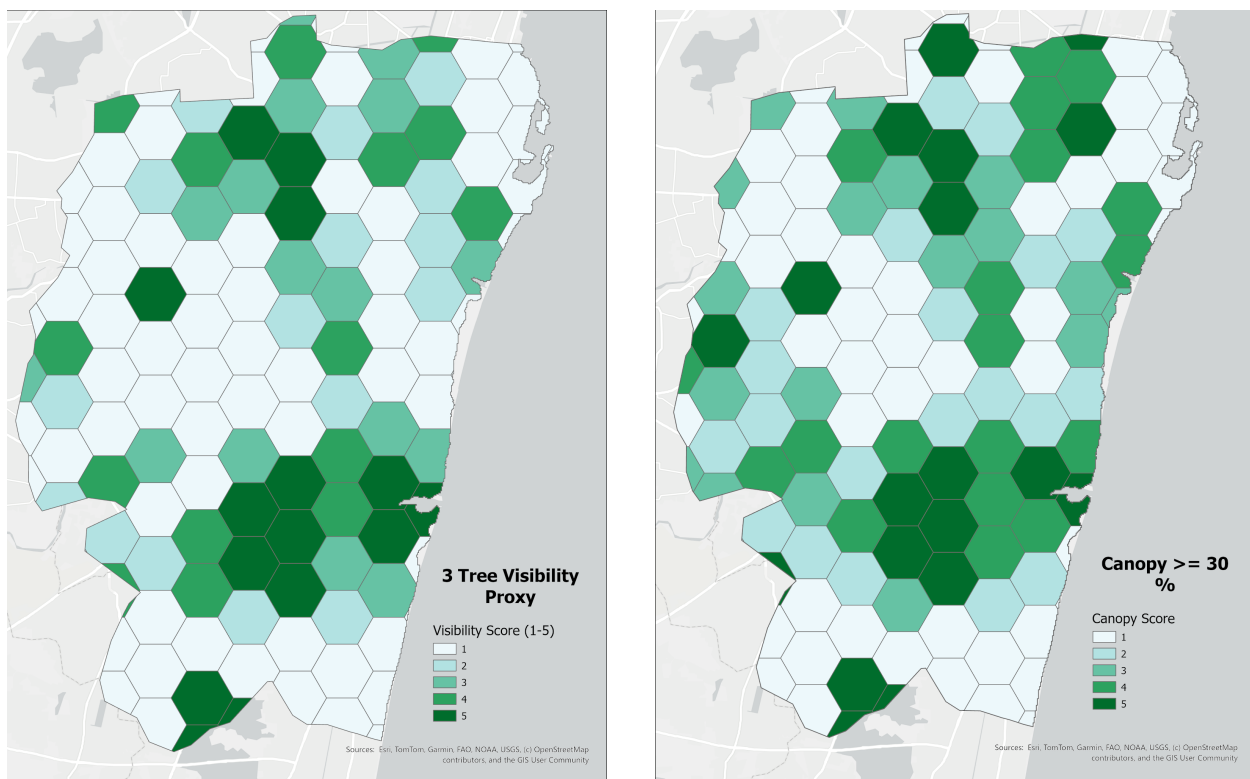
5.4 3-30-300

5.4.1 Three Trees Visible

The tree visibility indicator reveals severe deficits in clustered vegetation across Chennai. Scores ranged from 0.000 to 0.402, with a median of 0.000 and mean of 0.161. No neighbourhoods have a score greater than 0.500, indicating that none of the neighborhoods comprehensively meet the “three trees visible” proxy threshold. Spatial patterns show higher tree visibility concentrated in southern and specifically coastal hexagons and scattered pockets in the northern neighbourhoods, corresponding to an established national park (Guindy) and river bank (Kosasthalaiyar). The lowest scores are in what has now become central to the city in the western and northern focusses of this map, possibly as a result of urbanization and industrial expansion.

5.4.2 30% Canopy Cover

Canopy cover performance indicates widespread failure to meet the 30% standard across Chennai. Scores ranged from 0.000 to 0.980, with a median of 0.005 and mean of 0.030. Zero hexagons achieved scores at or above 1.0 (meeting the 30% canopy threshold). The NDVI threshold of 0.4 effectively isolated dense woody canopy from herbaceous vegetation, though the binary classification approach cannot distinguish canopy quality, species composition, or vertical structure.



(a) Spatial distribution of the tree visibility indicator across Chennai's hexagonal grid

(b) Spatial distribution of canopy cover scores relative to the 30% standard

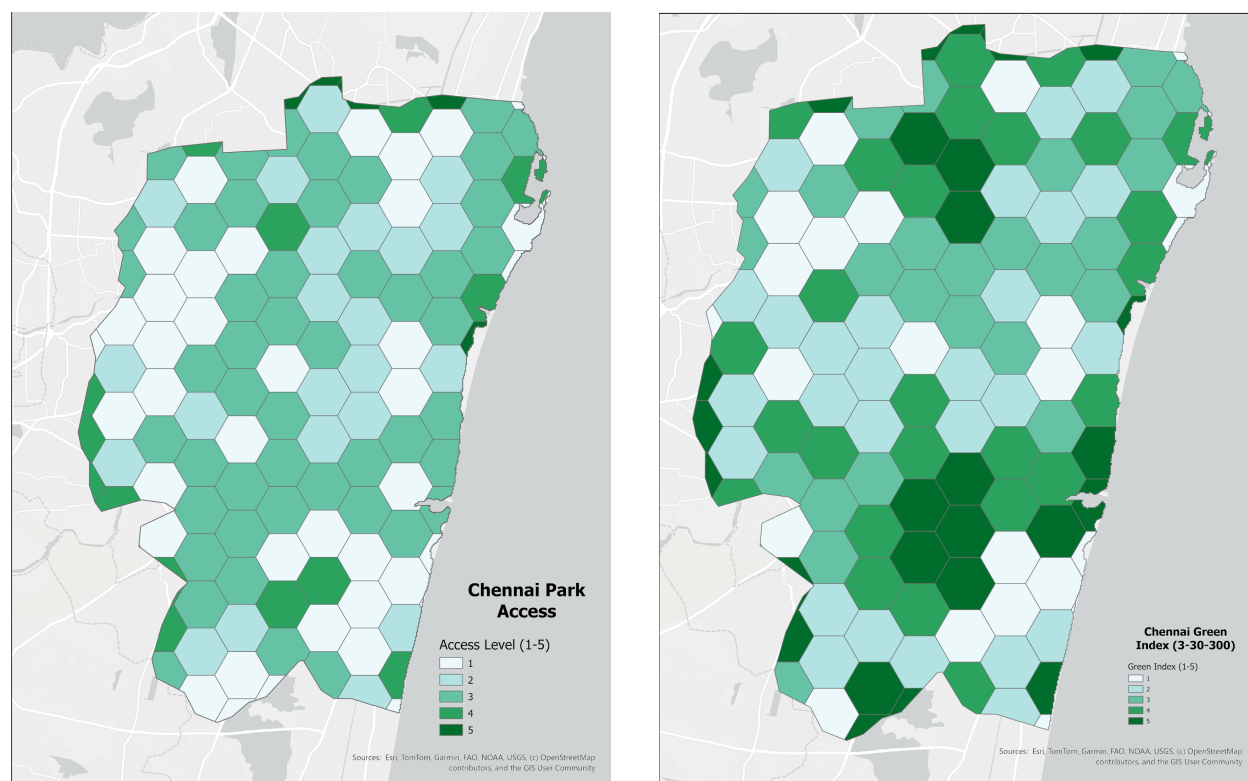
Figure 7: 3-30-300 green coverage indicators: tree visibility and canopy cover

5.4.3 300 m Park Access

Park accessibility shows highly uneven distribution across Chennai's hexagonal grid. Scores ranged from 0.000 to 1.014, with a median of 0.073 and mean of 0.100. One neighbourhood achieved full compliance (score = 1.0), indicating complete coverage within 300-meter network distance of parks. The graduated scoring approach captured partial access nuances - many neighbourhoods score between 0.200–0.600, indicating park proximity benefits a portion but not the entirety of residents.

5.4.4 Combined Greenness Index (3-30-300)

The integrated greenness index aggregates performance across all three indicators, revealing systemic green coverage inadequacy. Scores ranged from 0.000 to 0.461. Sixty-six neighbourhoods scored below a median threshold of 0.030, representing the bottom 50% identified as low-greenness areas requiring intervention. The equal-weighted averaging approach reveals that few hexagons perform well across all three dimensions - most exhibit mixed performance, scoring moderately on one or two indicators while failing others. This multidimensional assessment exposes neighborhoods that superficially appear adequate on single metrics but lack comprehensive green coverage.



(a) Park accessibility scores based on 300 m network service areas

(b) Combined greenness index (3-30-300) across Chennai's hexagonal grid

Figure 8: 3-30-300 green coverage indicators: park access and combined greenness index

5.5 Priority Zones

The intersection of low greenness areas (bottom 50%, $\text{greenness}_{333} < 0.030$) and high heat risk zones (top 25%, $\text{heat_risk} > 134.458$) identified 10 priority hexagons requiring urgent attention.

These zones represent spatial co-occurrence of green coverage deficit and elevated heat vulnerability, not a causal relationship. The 10 priority hexagons cover roughly 20 km² and affect an estimated over 866,000 residents. Geographically, priority zones concentrate in the north-central urban core and western periphery. These locations experience compounded risk: residents lack cooling infrastructure precisely where heat exposure and demographic susceptibility converge, creating conditions where green coverage interventions could yield maximum co-benefits for both thermal comfort and climate adaptation.

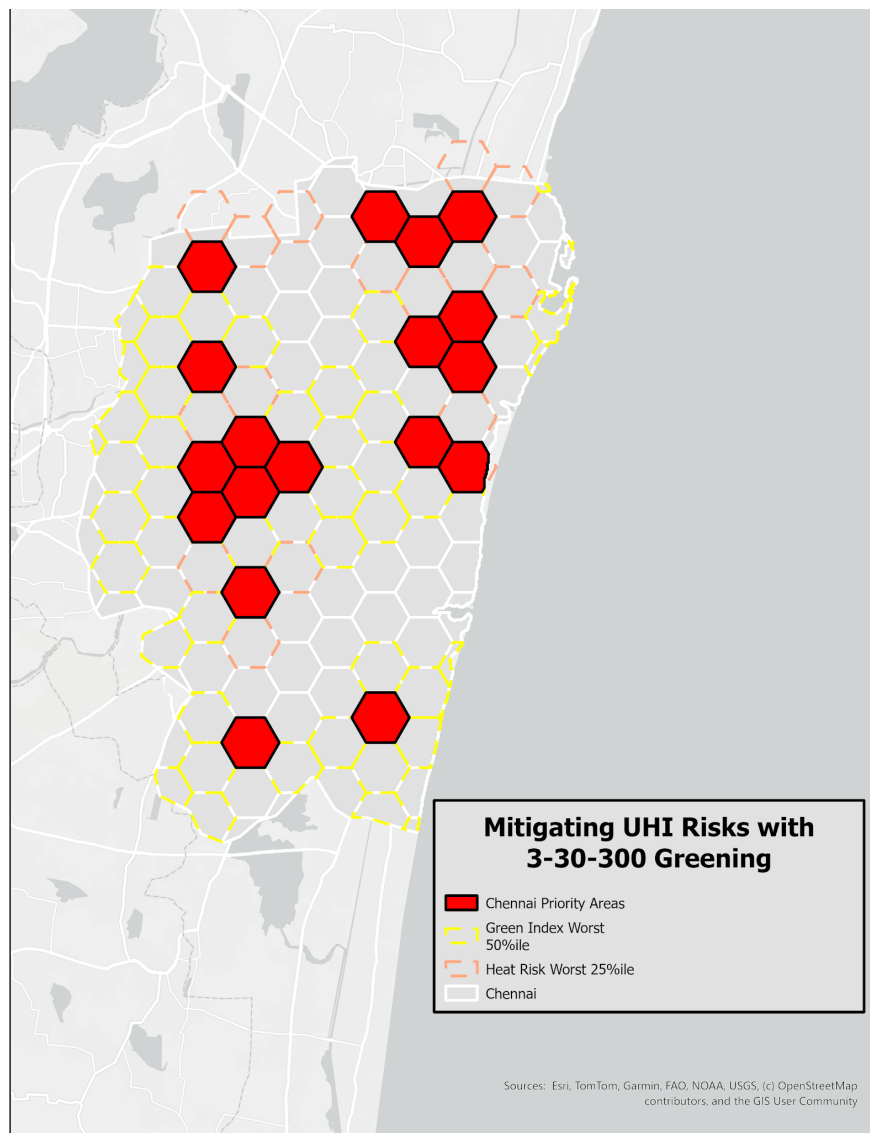


Figure 9: Priority intervention zones: hexagons with low greenness and high heat risk

6 Discussion

6.1 Key Findings

Three results from this analysis carry particular significance for Chennai's urban heat governance. First, the coastal sensitivity analysis produced a counterintuitive finding: geographic proximity to

the Bay of Bengal does not inherently guarantee lower land surface temperatures. Several coastal-adjacent areas exhibit slightly *higher* temperatures than inland locations. This is consistent with recent local assessments (TNGCC and ICLEI South Asia, 2026) and indicates that urban heat in Chennai is dictated primarily by localized land cover characteristics — dense built-up surfaces and limited vegetation — rather than by proximity to marine cooling. The practical consequence is that coastal wards cannot be assumed safe from heat stress, and the rural baseline for SUHII estimation should not rely on peri-urban zones that are themselves undergoing rapid land-cover change.

Second, while the physical heat hazard is pervasive across the metropolis, the underlying drivers of human risk are spatially segregated. Social vulnerability concentrates in the northern fringes, shaped by informal settlements and infrastructure deficits, whereas population exposure peaks in the historic urban core, driven by extreme density. This mismatch means that no single ward captures the full risk picture. By downscaling from ward-level averages to a 2 km hexagonal grid, the framework exposed micro-scale islands of compounded risk — pockets where extreme population density and deep socio-infrastructure disadvantages collide — that ward boundaries obscure. These localized pockets provide precise areas for intervention rather than broad administrative targets. Third, the 3-30-300 assessment revealed a systemic green coverage deficit: no neighbourhoods meet the tree-visibility proxy standard and only one approaches the 30% canopy target. No neighbourhood satisfies all three criteria. Paradoxically, this pervasiveness makes targeted action more tractable. By intersecting the bottom 50% of greenness scores with the top 25% of heat risk, the framework identified exactly 10 priority hexagons — covering roughly 20 km² and over an estimated 866,000 residents. Concentrating green coverage investment in these zones offers the highest marginal return for public health and climate adaptation. However, the case for green infrastructure cannot rest on heat reduction alone; Section 8 quantifies the co-benefits — stormwater interception, air quality improvement, carbon sequestration — that make this investment defensible across multiple policy domains.

6.2 Planning Implications

6.2.1 From Spatial Prioritisation to Implementation

The 10 priority zones provide an evidence base for targeted investment, but translating spatial analysis into on-ground greening requires navigating a complex stakeholder landscape. Implementation depends on buy-in from at least four actor groups: municipal corporations that control park budgets and street-tree planting programs; private landowners whose cooperation is essential for canopy expansion on residential and commercial parcels; community organisations representing the neighbourhoods prioritised for intervention; and funding agencies that require quantified return-on-investment projections. These governance realities — what blocks planting, what unlocks land, what persuades finance — are not hypothetical. Section 7 reports on a structured expert consultation with 8 practitioners across research, policy, and government that tested these assumptions against on-the-ground experience in Indian cities.

6.2.2 The Co-Benefit Value Proposition

A single-benefit framing — heat reduction alone — is insufficient to align these actors. Green infrastructure advocacy must articulate returns across stormwater management, air quality, property values, mental health, and biodiversity, so that multiple departments (health, environment, planning, stormwater) can each identify a stake in the same intervention. Holistic benefit measurement frameworks such as i-Tree ecosystem service valuation or natural capital accounting can monetise

avoided healthcare costs, reduced cooling energy demand, and postponed grey infrastructure investments, strengthening the economic case in fiscally constrained municipalities. Section 8 develops this argument in detail, presenting research-backed evidence across four co-benefit branches and quantifying projected returns for the 10 priority zones using the i-Tree Planting tool.

6.2.3 Participatory Governance

Finally, participatory planning processes that engage residents of priority zones in species selection, maintenance protocols, and design preferences build social license and ensure that interventions reflect community needs rather than impose top-down solutions. Without this multidimensional value proposition and inclusive governance, even the most rigorous spatial prioritisation risks remaining a planning exercise rather than catalysing tangible urban transformation.

6.3 Methodological Contributions and Transferability

Two design choices distinguish this framework from prior Chennai heat assessments and enhance its transferability to other cities.

6.3.1 Halo-Based SUHII Estimation

Prior studies on urban heat in Chennai often adopt an in-city comparison approach, identifying relatively cooler zones within the city to serve as the rural baseline ([TNGCC and ICLEI South Asia, 2026](#)). That approach captures intra-urban variation and is useful for neighbourhood-level planning. However, in rapidly urbanizing contexts like Chennai, areas classified as "rural" within the city may no longer represent true baseline conditions due to ongoing land-use change.

This study defines the rural baseline using a halo buffer outside the urban boundary, providing an external and standardised reference independent of intra-city land cover variation. Combined with a hexagonal tessellation that ensures consistent spatial units, this design reduces sensitivity to irregular administrative boundaries and enhances cross-city comparability. The consistency of findings across both approaches — particularly regarding the limited cooling effect of coastal proximity — strengthens confidence in the robustness of the SUHII estimates.

6.3.2 Open-Data 3-30-300 Assessment

The 3-30-300 assessment relies entirely on freely available satellite imagery (Landsat NDVI; [U.S. Geological Survey, 2024](#)), OpenStreetMap road networks, and municipal park inventories — eliminating dependency on proprietary data or field surveys. Any city with Landsat coverage can replicate the tree-visibility and canopy-cover indicators; cities with documented park locations and road centrelines can implement the full framework. The hexagonal tessellation provides a standardised spatial unit for cross-city comparison, and the equal-weighting integration formula ensures methodological consistency across geographies. This workflow operates within the technical capacity of municipal planning departments equipped with ArcGIS Pro or equivalent open-source platforms, lowering barriers to adoption in cities where urbanisation outpaces environmental planning capacity.

7 User Research — Expert Consultation Summary

In parallel with the spatial and co-benefit analysis, a structured expert consultation was conducted over five weeks to scope the solution space and validate key methodological assumptions with

practitioners. Twenty-two domain experts working at the intersection of urban heat and India were identified; eight engaged across live interviews and asynchronous sessions. Experts were segmented into three personas - Thinkers, Enablers, and Financiers - each probed on a tailored but overlapping question set. The full synthesis is available as a companion Knowledge Repository. This section presents the process, the thematic clustering, and the findings most directly relevant to the co-benefit analysis that follows.

7.1 Research Process

Table 7: Expert consultation: outreach pool, conversion rates, and persona definitions.

Persona	Pool	Engaged	Definition
Thinker	9	3	Researchers & academics: framework robustness, data needs, tool design.
Enabler	7	3	International org & NGO implementers: policy translation, governance, barriers.
Financier	5	2	Government officials & urban practitioners: ROI, feasibility, financial structures.
Total	21	8	Overall conversion $\approx 38\%$; 5-week fieldwork period.

7.2 Thematic Clustering Matrix

Seven themes emerged across the eight conversations. Table 8 shows how many experts from each persona contributed to each theme (T / E / F) and the core insight synthesised across responses.

7.3 Thematic Synthesis

The seven themes are summarised below with emphasis on findings that directly inform the methodology and recommendations of this report. Full interview notes and per-theme attribution are available in the companion Knowledge Repository.

7.3.1 Data, Tools & Methodological Robustness (Themes 1 & 2)

The most consistent methodological feedback was the need to move beyond single-date land surface temperature analysis toward multi-temporal and higher-resolution approaches. Two independent Enabler-persona experts - Rahul Agnihotri (UNEP) and Leigh Mante (ORF Middle East) - flagged that temperature alone significantly underestimates heat stress in coastal cities; relative humidity must be incorporated, particularly in Chennai’s Bay of Bengal context. This finding informed the methodological caution noted in Section 4.2 regarding SUHII as a partial proxy for physiological heat stress.

On tools, a clear consensus emerged across all three personas: an interactive, scenario-aware dashboard is necessary for genuine decision-making, while static StoryMaps serve only communication purposes. Critically, tools without accompanying capacity-building workshops do not get embedded into governance workflows - training is not an add-on but part of the deliverable. The equity dimension of access to cooling infrastructure (e.g., AC) was flagged as a risk layer the dashboard should surface explicitly alongside the Social Vulnerability Index.

7.3.2 Implementation Barriers & Governance (Themes 3 & 4)

Prof. Rutul Joshi (CEPT University) provided the most granular account of on-ground barriers to green-coverage interventions in Indian cities. The primary constraint on public-land planting is

Table 8: Thematic clustering matrix across expert personas.

#	Theme	T	E	F	N	Core Insight
1	Data Gaps & Methodological Robustness	3	2	0	5	Multi-temporal LST and Sentinel-2 imagery needed for seasonal validation. Humidity - not temperature alone - is the correct heat-stress indicator for coastal cities.
2	Tool Usability & Decision-Support Design	3	2	1	6	Interactive dashboards outperform static StoryMaps. Tools must ship with capacity-building workshops.
3	Implementation Barriers - Physical & Regulatory	1	3	1	5	Road-expansion reservations block public-land planting. Private parcels are the scalable frontier via Green FAR mechanisms.
4	Governance & Cross-Departmental Coordination	0	3	2	5	Siloed departments share benefits but not budgets. Legal mandates or direct ward-budget linkages are the most reliable coordination mechanisms.
5	Co-Benefit Quantification & the ROI Frame	1	2	2	5	Co-benefits are universally acknowledged but not systematically measured. Health outcomes drive government action most strongly.
6	Local Context & Chennai-Specific Realities	2	2	1	5	Coastal geography, water-body freeze rules, the 10% green-area mandate, and ward governance require bespoke adaptation.
7	Geographic Transferability & Framework Scalability	2	2	0	4	High relevance for Gulf and South Asian coastal cities. Adaptations needed: humidity weighting, cooling degree days, mangrove co-benefits.

T = Thinker count; E = Enabler count; F = Financier count; N = total. Experts may contribute to multiple themes.

not zoning regulation but road-widening corridor reservations: planners refuse to plant trees that will be removed within five to ten years. This makes private parcels the more scalable opportunity - unlockable through a Green Floor Area Ratio (Green FAR) mechanism embedded in building-permission approvals, or property tax abatements linked to satellite-verified canopy maintenance.

On governance, Rahul Agnihotri (UNEP) and Joshi independently converged on the same diagnosis: greening interventions span multiple departments (health, environment, planning, stormwater) that share benefits but not budgets. Formal inter-departmental structures - steering committees, climate cells - are necessary but rarely created proactively. Joshi was emphatic on a specific Indian institutional reality: the most reliable coordination mechanisms are either legally mandated or directly linked to financial flows. Voluntary cooperation is structurally fragile. Ex-Secretary Anshul Mishra (CMDA) reinforced this from an insider's perspective, adding that any solution must be grounded in existing law and regulation to be credible to executive agencies.

7.3.3 Co-Benefit Framing & the ROI Imperative (Theme 5)

This theme most directly validates the co-benefit analysis in Section 8. Agnihotri confirmed that co-benefits including health, income, and well-being improvements are recognised within UNEP and the World Bank, but no comprehensive study currently measures their magnitude systematically - the gap that Section 8 attempts to close. Health-related outcomes were identified as the single most persuasive co-benefit for government action, particularly for vulnerable and informal-settlement populations. This finding motivated the decision to foreground Branch A (temperature reduction and mortality) in the CBA framework rather than leading with flood reduction or air quality.

Leigh Mante (ORF Middle East) provided the clearest articulation of the ROI framing imperative: investment decisions are driven by return on investment, not environmental urgency. Framing green coverage as a bankable sustainable-cooling investment - rather than an environmental obligation - is consistently more persuasive to both government and finance audiences. This directly motivates the 'ranking flip' methodology: demonstrating that tree planting becomes a substantially stronger investment when all co-benefit branches are counted together, not just thermal reduction alone. Romit Sinha (Urban Practitioner) endorsed this framing from the implementation side, noting strong current momentum around heat risk policy in India as a favourable window.

7.3.4 Local Context & Transferability (Themes 6 & 7)

A cross-cutting signal from all India-experienced experts was the imperative to avoid over-indexing on UHI as a singular priority. Urban administrations manage competing demands - infrastructure, sanitation, economic development - and a framework that positions heat risk as uniquely paramount loses credibility with implementers. The co-benefit approach responds to this directly: by demonstrating that a single intervention generates returns across multiple urban priority domains simultaneously, it makes a more defensible claim on constrained resources. Mishra additionally noted that Chennai's existing blue-green policy instruments (water-body freeze rule, 10% green mandate in new developments) are mobilisable assets that should be positioned as a foundation rather than ignored.

Sinha proposed stress-testing recommendations across distinct 2 sq. km hexagon types within Chennai - industrial, residential, coastal, peri-urban - to expose how local realities vary and harden the framework against failure modes. On transferability, Mante confirmed high applicability to Gulf coastal cities (Dubai, Abu Dhabi) with additions of humidity weighting and mangrove co-benefits,

while Intikhab Alam (GIS Analyst, Pakistan) confirmed relevance to South Asian cities including Karachi and Lahore, with adaptations needed for census variable localisation and open-source implementation pathways.

7.4 Validated Assumptions & Unresolved Tensions

Table 9: Validated assumptions and unresolved tensions from expert consultation.

Validated by Expert Consultation	Unresolved Tensions
Multiplicative risk score (hazard $\times$ vulnerability $\times$ exposure) is consistent with World Bank practice.	Humidity vs. temperature: incorporating humidity would require additional datasets and a revised risk-score formulation.
Green coverage is feasible within current Chennai regulatory schemes - no new legislation required to begin.	Proxy reliability: greening and flood risk reduction is directionally confirmed but uncertain in magnitude for extreme storm events.
Co-benefits across health, flooding, and air quality are real; the gap is quantification, not existence.	Maintenance gap: co-benefits rarely materialise without institutionalised maintenance; the CBA may overstate long-run returns.
The ‘ranking flip’ - co-benefits change the investment ranking in favour of nature-based solutions—is recognised and compelling.	Economic growth competition: UHI reduction may compete with industrialisation as a policy priority.
Spatial prioritisation frameworks support coordination when paired with formal governance structures. India’s current heat risk momentum creates a genuine policy window.	

7.5 Practitioner Voices

“Co-benefits such as improved air quality, increased oxygen levels, and better health outcomes are widely expected, especially for outdoor workers. However, these benefits are generally not measured or monitored, making it difficult to confirm their magnitude in practice.”

—Rahul Agnihotri, Senior Programme Officer, UNEP (Enabler)

“Analytical tools alone are not enough to attract funding. Investment decisions are ultimately driven by return on investment. Framing urban heat as an economic opportunity rather than purely an environmental issue is more effective.”

—Leigh Mante, Junior Fellow, Climate & Energy, ORF Middle East (Enabler)

7.6 Implications for the Co-Benefits Analysis

The expert consultation shapes Section 8 in four concrete ways. First, the co-benefit framework foregrounds Branch A (temperature reduction $\rightarrow$ health outcomes) as the primary policy lever, following the consensus that health benefits are the single most persuasive argument for government action. Second, the CBA is structured to demonstrate the ‘ranking flip’ explicitly - presenting results first with Branch A alone, then with all branches combined - because experts confirmed this comparative framing is the most compelling to financier-persona decision-makers. Third, the

discount rate and benefit-attribution model follows the ward-level and departmental structure recommended by Joshi and Agnihotri, so that it maps onto existing government budget categories rather than remaining abstract. Fourth, the unresolved tension around maintenance costs is acknowledged as a caveat in the CBA’s long-run projections.

The full expert consultation, individual interview notes, and thematic deep-dive are available in the companion document: *User Research Knowledge Repository* (Chen, Doshi, Masood, Zhou - Carnegie Mellon University / World Bank Group, May 2026).

8 Co-Benefits Analysis

8.1 Introduction

This section presents research-backed evidence on four co-benefit branches of urban tree canopy expansion: temperature reduction, rainfall interception, air filtering, and additional social, ecological, and economic co-benefits. Co-benefits are important in this case because urban tree canopy can address several urban challenges at once. In Chennai, the same intervention can reduce heat, lower runoff, improve air quality, and improve urban livability. Each finding is based on cited papers from our literature review or on peer-reviewed studies identified through systematic search. The research findings are general in scope. Each section also includes a Chennai-specific context area to show how the evidence applies locally. The section also draws on the i-Tree Planting tool ([USDA Forest Service, 2024](#)) to quantify co-benefits for the 10 priority intervention zones identified in our GIS analysis. The paper focuses on the 30% tree canopy target from the 3-30-300 framework and applies this target to Chennai, India.

8.2 Branch A: Temperature Reduction

8.2.1 How Trees Reduce Temperature

Tree canopy lowers urban temperatures through shading (blocking solar radiation from reaching surfaces) and evapotranspiration (releasing water vapor that cools surrounding air).

8.2.2 Research Findings: Canopy and Temperature

A 10% increase in tree canopy reduces urban air temperature by 0.8°C. A 30% increase lowers air temperature by up to 1.5°C ([Ziter et al., 2025](#)). Tree canopy cover is the dominant cooling factor, explaining 67% of the spatial variation in urban air temperature ([Ziter et al., 2025](#)). In Phoenix residential neighborhoods, increasing canopy from 10% to 25% produced up to 2.0°C average daytime cooling at the local scale ([Middel et al., 2014](#)). Cooling increases nonlinearly as canopy cover rises, with the strongest additional cooling occurring once canopy cover exceeds about 40%, especially at the city-block scale ([Ziter et al., 2019](#)). Urban tree canopy mitigates approximately 48.6% of the urban heat island effect globally on average ([Li et al., 2025](#)).

8.2.3 Research Findings: Temperature and Health

Each 0.1 increase in NDVI (a measure of vegetation greenness) reduces heat-related mortality by 6–10% in low-GDP regions ([Ballester et al., 2025](#)). Epidemiological time-series studies for India provide exposure-response functions linking heat stress to mortality, reported as relative risk (RR) and attributable fraction, stratified by climate zone and population group ([de Bont et al., 2024](#)).

8.2.4 Chennai Context

Chennai's mean Land Surface Temperature rose from 29.4°C in 2000 to 33.6°C in 2020, a 4.2°C increase over two decades. Maximum and minimum temperatures are expected to increase by 2.9°C and 3.3°C respectively by the end of the century (TNGCC and ICLEI South Asia, 2026). The number of annual heatwave days in Chennai is projected to increase from 42 to 81 by 2050. Tamil Nadu's average annual heatwave days doubled from 15 to 30 between 2011 and 2019, and the state's maximum temperature has increased by 1.5°C since 1981. In 2023, the state reported 12 heatwave-related deaths (TNGCC and ICLEI South Asia, 2026). A clear negative correlation exists between LST and NDVI in Chennai: increased vegetation cover is associated with lower surface temperatures (TNGCC and ICLEI South Asia, 2026). 25% of Chennai's wards (50 wards) have a very high heat risk index, accounting for 30% of the city's total population but covering only 12% of the total city area (TNGCC and ICLEI South Asia, 2026). Chennai's per capita green cover is 8.75 sq.m, below the WHO standard of 9.5 sq.m per person (TNGCC and ICLEI South Asia, 2026). Our team's SUHII analysis measured Chennai's urban heat island intensity at 5.024°C (Chen et al., 2026b). Built-up land in Chennai's periphery surged from 11.9% to 44.16%, intensifying heat exposure (Muthiah et al., 2022). For India, the recommended social discount rate is 8% for general projects and 6% for long-life environmental and infrastructure projects (Institute of Economic Growth, 2023).

8.3 Branch B: Rainfall Interception

8.3.1 How Trees Reduce Flooding

Tree canopy intercepts rainfall before it reaches the ground. Leaves and branches catch water that then evaporates instead of becoming surface runoff. Root systems promote soil infiltration. Both reduce stormwater volume entering drainage systems. The same canopy expansion that cools a neighborhood also reduces flood risk.

8.3.2 Research Findings: How Much Rainfall Do Trees Intercept?

Individual urban trees intercept between 15% and 28% of rainfall falling on their canopy area. Depending on species, canopies can retain between 0.03 and 2.24 mm of water per square meter of leaf area (Kuehler et al., 2017; Center for Watershed Protection, 2016). In closed-canopy forests, interception loss is approximately 18–29% for hardwood forests, and approximately 18–45% for coniferous forests (Carlyle-Moses and Gash, 2011). Broadleaf evergreen and conifer stands intercept 36% of summer rainfall (LAI = 6.1). Mixed conifer and deciduous stands intercept 18% (LAI = 3.7) (Xiao and McPherson, 1998). Individual street trees intercept 610 to 26,000 liters of annual rainfall per tree, varying by species and tree characteristics (Xiao and McPherson, 2002). Urban birch trees intercept 25.6% of gross rainfall on average. Canopy attenuates average rainfall intensity by 50.2% during the leafed season and 41.6% during the leafless season (Zabret and Šraj, 2023).

8.3.3 Research Findings: How Much Does This Reduce Runoff?

Urban trees can reduce stormwater runoff volume by 20–25% and peak flow by 16–25%. Infiltration and storage beneath tree canopies account for over 20% of the water balance (Gonzalez-Casares et al., 2025). Removing 31 mature street trees from a residential catchment increased surface runoff by 198 cubic meters (4% of total runoff). Per tree, 66 liters of rainfall per square meter of canopy was lost that would otherwise have been intercepted (Selbig et al., 2022). Even during intense, high-volume storm events, trees still provide up to 13.2% reduction in runoff (Gonzalez-Casares

et al., 2025). Small storms are responsible for most pollutant washout. Urban forests produce greater benefits through water quality protection than through flood control for large events (Xiao and McPherson, 1998).

8.3.4 Chennai Flood Context

Chennai receives approximately 1,300 mm of rainfall annually. Recent years have seen intense downpours over short periods leading to urban flooding. Annual rainfall is expected to decrease by 5% by 2050, but the intensity of individual events is increasing (TNGCC and ICLEI South Asia, 2026). Wards 32, 44, 76, and 78 in Chennai are identified for bioswales, rain gardens, and waterbody restoration because they face both high heat risk and frequent waterlogging (TNGCC and ICLEI South Asia, 2026). Urban growth and sprawl in Chennai directly increase flood depth, extent, and inundation area under future development scenarios (Nithila Devi et al., 2019). When stormwater co-benefits are excluded from CBA, grey infrastructure dominates. When included, green-grey and green-blue-grey portfolios become economically viable (Alves et al., 2019; Ruangpan et al., 2024).

8.4 Branch C: Air Filtering

8.4.1 How Trees Clean Air

Tree leaves physically capture particulate matter (PM_{2.5}, PM₁₀) on their surfaces. They absorb gaseous pollutants (NO₂, SO₂, O₃) through stomata. Once inside the leaf, gases diffuse into intercellular spaces and are permanently converted. Trees also sequester carbon dioxide, storing it as biomass.

8.4.2 Research Findings: How Much Pollution Do Trees Remove?

Urban trees remove an average of 6.73 grams of pollutants per square meter of canopy cover per year across all pollutant types (Nowak et al., 2014). In Strasbourg, France, city-managed trees removed approximately 88 tonnes of pollutants in one year: 1 tonne of CO, 14 tonnes of NO₂, 56 tonnes of O₃, 12 tonnes of PM₁₀, 5 tonnes of PM_{2.5}, and 1 tonne of SO₂ (Selmi et al., 2016). Across the United States, urban trees remove an estimated 711,000 tonnes of particulate matter per year (Nowak et al., 2014).

8.4.3 Research Findings: Health Impact of Air Pollution Removal

PM_{2.5} removal by urban trees ranged from 4.7 tonnes per year in Syracuse to 64.5 tonnes per year in Atlanta. The monetary value of this removal ranged from \$1.1 million per year (Syracuse) to \$60.1 million per year (New York City). Most of this value came from reducing human mortality (Nowak et al., 2013). A 10 $\mu\text{g}/\text{m}^3$ increase in PM_{2.5} is associated with approximately 4% increased risk in all-cause mortality, 6% in cardiopulmonary mortality, and 8% in lung cancer mortality (Pope et al., 2002).

8.4.4 Chennai Context

Energy consumption in Chennai's buildings sector contributed 71% of total GHG emissions, mainly due to reliance on thermal power for urban cooling (TNGCC and ICLEI South Asia, 2026). Street tree planting can reduce cooling loads in adjacent buildings by 5–10%. Vertical gardens and green walls can reduce building cooling loads by 5–15% (Chen et al., 2026a). Chennai faces serious

air quality challenges, particularly with PM_{2.5} and NO₂. Tree canopy expansion in priority tessellations would concentrate filtration benefits in neighborhoods with the highest combined heat exposure and social vulnerability.

8.5 Branch D: Additional Co-Benefits

Beyond temperature reduction, rainfall interception, and air filtering, urban tree canopy expansion generates several additional co-benefits that are documented in the literature and relevant to Chennai.

8.5.1 Public Health and Mental Wellbeing

Tree planting in high-risk Chennai wards (54–57) directly benefits slum communities along riverbanks and canals who are highly vulnerable to urban heat. Pocket parks and community gardens provide heat relief, mental health benefits, community cohesion, and opportunities for physical activity (Chen et al., 2026a; TNGCC and ICLEI South Asia, 2026). Wards 44, 48, 52, 55, 73, and 75 together contain 45 schools, 11 markets, and 35 bus stops. Shading structures and tree planting in these public spaces would improve thermal comfort for commuters, pedestrians, and children (TNGCC and ICLEI South Asia, 2026). Fourteen wards in Chennai have very high vulnerability due to higher slum populations, inadequate road infrastructure, low NDVI, fewer parks, and limited medical facilities (TNGCC and ICLEI South Asia, 2026).

8.5.2 Biodiversity and Ecology

Street tree planting and micro forests create habitat for urban birds and insects, improving urban biodiversity. The Tamil Nadu government’s Karunkadugal initiative promotes development of mini forests in cities using dense, multi-species native plantations (TNGCC and ICLEI South Asia, 2026). Waterbody restoration in wards 32, 44, 76, and 78 supports wetland habitat restoration and aquatic biodiversity alongside heat and flood risk reduction (Chen et al., 2026a; TNGCC and ICLEI South Asia, 2026).

8.5.3 Economic Benefits and Employment

Tree planting and urban greening create jobs in nurseries, planting and maintenance, urban forestry, ecological restoration, and park management. MSME clusters in Chennai can benefit from cooling interventions including landscaping, energy-efficient cooling appliances, and solar rooftops (Chen et al., 2026a; TNGCC and ICLEI South Asia, 2026). Cool roofs can reduce building cooling energy demand by 10–30% (Chen et al., 2026a; TNGCC and ICLEI South Asia, 2026).

8.6 Connecting the Four Branches

8.6.1 Why Co-Benefits Must Be Included in Appraisal

Tree planting programs are often seen as weak investments. When cities evaluate them purely on cooling or purely on aesthetics, the costs of planting and decades of maintenance can outweigh the measured returns. This leads decision-makers to favor grey infrastructure like concrete drainage or engineered cooling systems, which show clearer single-purpose returns.

But this framing is flawed. It measures one benefit and ignores the rest. As this paper has shown, the same trees that reduce temperature also intercept rainfall, filter air pollution, store carbon,

improve public health, and support biodiversity. When appraisals only count one of these, they systematically undervalue green coverage.

The research confirms this: when co-benefits are excluded from cost-benefit analysis, grey infrastructure dominates the feasibility rankings. When co-benefits are included (energy savings, air pollution reduction, carbon sequestration, thermal comfort valued through willingness-to-pay), combined green-grey and green-blue-grey solutions become economically viable (Alves et al., 2019). NBS strategies achieved a Benefit-Cost Ratio above 1 in three of four case study cities over a 30-year life cycle (Ruangpan et al., 2024). CBA outcomes depend strongly on which benefits are counted. Studies that include a broader set of co-benefits consistently find stronger results for green infrastructure (Chelli et al., 2024).

This is the ranking flip. Tree planting doesn't become a better investment when co-benefits are added. It was always that good. Traditional appraisal just wasn't counting properly.

8.6.2 Recommended Frameworks for Chennai

Several established frameworks exist for evaluating co-benefits. Life-Cycle Cost-Benefit Analysis (LCCBA) lays out all benefits and costs year by year over the project's lifetime, discounts future values to the present, and produces standard investment metrics like NPV and BCR. Ruangpan et al. used this approach with a 30-year horizon and a 3% discount rate across four cities (Ruangpan et al., 2024). Multi-Criteria Decision Analysis (MCDA) scores interventions across multiple objectives and lets stakeholders assign weights based on local priorities. Alves et al. used this to rank green and grey infrastructure across flood reduction, cost, and co-benefits (Alves et al., 2018). InVEST (Integrated Valuation of Ecosystem Services and Tradeoffs) models how much ecosystem service an intervention generates under different scenarios. Biasin et al. used it in Turin to evaluate NBS scenarios addressing both UHI and flooding simultaneously (Biasin et al., 2023).

For Chennai, we recommend MCDA as the primary framework, because it can accommodate both monetizable benefits (energy savings, avoided flood damage) and the harder-to-quantify benefits (equity, biodiversity, public health) that matter to the World Bank and local stakeholders.

As a practical extension, we are using the **i-Tree Planting tool** to generate the physical co-benefit quantities (stormwater intercepted, CO₂ stored, pollutants removed, energy saved) for priority zones in our GIS analysis. These i-Tree outputs feed into the broader MCDA evaluation as quantitative inputs, re-priced for Chennai's local costs.

8.7 Summary of Key Findings

See Table 10 for a summary of key findings across all four co-benefit branches.

8.8 i-Tree Planting Tool: Quantifying Co-Benefits for Chennai

8.8.1 What is the i-Tree Planting Tool?

i-Tree Planting is a free web-based tool developed by the USDA Forest Service that estimates the environmental co-benefits of tree planting projects over a defined time horizon. Given inputs including species, tree size at planting, location, and project duration, the tool forecasts cumulative benefits across five categories: carbon sequestration, stormwater interception, avoided runoff, air pollutant removal, and building energy savings (USDA Forest Service, 2024). For this project, i-Tree Planting was used to generate physical co-benefit quantities for each of the 10 priority hexagon

Table 10: Summary of key co-benefit findings by branch.

Branch	Finding	Metric	Source
A	10% canopy increase $\rightarrow$ -0.8°C air temp	0.8°C per 10%	Ziter et al., 2025
A	30% canopy $\rightarrow$ up to -1.5°C	1.5°C per 30%	Ziter et al., 2025
A	0.1 NDVI $\rightarrow$ 6–10% $\downarrow$ heat mortality (low-GDP)	6–10% per 0.1 NDVI	Springer, 2025
B	Trees intercept 15–36% of rainfall	15–36% interception	Kuehler+; Xiao+, 1998
B	Runoff $\downarrow$ 20–25%; peak flow $\downarrow$ 16–25%	20–25% reduction	Gonzalez-Casares+, 2025
B	Individual trees: 610–26,000 L/tree/yr	L/tree/yr	Xiao+, 2002
C	6.73 g pollutants/ m^2 canopy/yr	g/ m^2 /yr	Nowak et al., 2014
C	PM2.5 removal \$1.1M–\$60.1M/yr per city	\$/yr	Nowak et al., 2013
A	Chennai LST $+4.2^{\circ}\text{C}$ (2000–2020)	4.2°C increase	TNGCC, 2026
A	50 wards very high heat risk (30% of pop.)	50 wards	TNGCC, 2026
B	Wards 32,44,76,78: dual heat+flood risk	dual-risk wards	TNGCC, 2026
C	Building energy = 71% of GHG; trees $\downarrow$ 5–10%	71% emissions	TNGCC, 2026
D	Cool roofs $\downarrow$ cooling demand 10–30%	10–30% savings	CoBenefits Matrix
All	Standard CBA undervalues low-income benefits	Equity gap	Kind et al., 2020
All	India discount rate: 6% for env. infrastructure	6% base case	IEG, 2023

zones identified in the GIS analysis. A full methodological account is available in the standalone co-benefits analysis white paper.

8.8.2 Configuration for Chennai

Since Chennai is not available as a location in i-Tree, US National Average was selected following standard practice for non-US projects. Key parameters were localised as shown in Table 11.

Table 11: i-Tree Planting tool configuration parameters for Chennai.

Parameter	Value	Source
Electricity Emissions Factor	710 kg CO ₂ eq/MWh	CEA, 2024
Fuel Emissions Factor	63 kg CO ₂ eq/MMBtu	IPCC, 2006
Project Duration	40 years	Standard practice
Tree Mortality	3% per year (Annual)	Hilbert et al. (2019)
Species	<i>Albizia saman</i> (Rain Tree)	Selected for Chennai

Albizia saman was selected for its large mature canopy spread of 15 to 25 metres, fast growth rate, and proven tolerance of Chennai’s tropical heat, humidity, and monsoon conditions.

8.8.3 Number of Trees and Group Structure

The number of trees required per zone was calculated from each hexagon’s canopy gap relative to the 30% target, applied over a hexagon area of 2 km², with 200 m² assigned as the conservative mature canopy area per *Albizia saman*. Five zones with 0% existing canopy were grouped together in i-Tree, resulting in 6 input groups representing all 10 zones. Total trees across all 10 zones: **29,778**.

8.8.4 Results and Interpretation

Over the 40-year project period, the 10 priority zones are projected to deliver the co-benefits shown in Table 12.

Table 12: Projected 40-year co-benefits across 10 priority intervention zones.

Co-Benefit	Total Across 10 Zones
CO ₂ Sequestered	3,397 tonnes
Avoided Stormwater Runoff	49,307 m ³
Rainfall Interception	~860,000 m ³
O ₃ Removed	~74,000 kg
NO ₂ Removed	~6,150 kg
SO ₂ Removed	~13,270 kg
PM2.5 Removed	~3,890 kg

Carbon sequestration of 3,397 tonnes over 40 years reflects the slow but sustained biomass accumulation of *Albizia saman* at scale. At an average of around 340 tonnes per zone, this is consistent with Nowak et al. (2014) estimates of 6.73 grams of pollutants removed per square metre of canopy per year when applied across nearly 30,000 trees over four decades. These sequestration values should be read alongside the broader carbon context in Section 8.4, where building energy consumption accounts for 71% of Chennai’s total GHG emissions (TNGCC and ICLEI South Asia, 2026). Tree planting alone will not close that gap, but it contributes meaningfully and delivers co-benefits that energy efficiency measures do not.

Stormwater avoided runoff of 49,307 m³ is particularly relevant to Chennai’s flood risk profile. This aligns with Branch B findings that urban trees reduce runoff volume by 20 to 25% (Gonzalez-Casares et al., 2025), and is especially significant for wards 32, 44, 76, and 78, which face both high heat risk and frequent waterlogging (TNGCC and ICLEI South Asia, 2026).

PM2.5 removal carries the greatest health significance of the air quality results. As noted in Section 8.4, a 10 µg/m³ increase in PM2.5 is associated with a 4% increase in all-cause mortality (Pope et al., 2002). The priority zones were selected in part for their high social vulnerability scores, meaning these gains will benefit residents least able to protect themselves from pollution exposure.

Energy savings returned zero across all groups due to the tool’s US-based location setting and the exclusion of building climate controls, which do not apply to Chennai’s tropical context. This benefit category should be estimated separately using Chennai-specific building energy models.

A full breakdown of results by group and zone, along with detailed methodology and limitation discussion, is available in the standalone co-benefits analysis white paper.

9 Limitations and Future Work

9.1 Data Limitations

The social vulnerability component relies on the Census of India (2011), introducing a 14-year lag between the demographic data and the current analysis. Ward-level socio-economic conditions — particularly informal settlement boundaries, infrastructure quality, and population density —

may have shifted substantially over this period due to urbanisation, redevelopment, and migration. Until the next national census is released, this temporal gap remains an unavoidable constraint for any ward-level vulnerability assessment in Indian cities.

A related caveat concerns administrative boundaries. The Chennai boundary used in this project reflects older delineations that have since been updated. While the underlying framework for calculating urban heat risk and mapping green coverage does not depend solely on this administrative boundary and will persist under newer delineations, relatively old data and boundaries may have understated the extent of heat risk. This boundary was selected for maximum data availability, given that the 2011 Census data was mapped to these older administrative units. The 3-30-300 indicators inherit the spatial resolution of Landsat imagery (30 m), which cannot detect small green spaces below approximately 900 m² and cannot distinguish tree canopy beneath building overhangs. The binary NDVI thresholds (0.35 for tree visibility, 0.4 for canopy cover) are effective proxies for the presence of woody vegetation but do not capture canopy quality, species composition, vertical structure, or seasonal phenology beyond the summer acquisition date used. The parks dataset (629 locations) records point centroids and approximate size categories but does not capture park quality, maintenance status, amenities, or accessibility barriers such as fences, gates, and operating hours. The OpenStreetMap road network used for the 300 m service-area analysis represents vehicular infrastructure more completely than pedestrian routes; informal paths and shortcuts that residents actually use for park access are not documented.

9.2 Methodological Limits

SUHII captures the thermal hazard dimension of heat risk but does not incorporate relative humidity, which multiple expert consultees identified as the more physiologically relevant heat-stress indicator for coastal cities like Chennai. The composite risk score therefore underestimates physiological heat stress in high-humidity areas near the coast and along waterways. The multiplicative risk formulation ($\text{SUHII} \times \text{Vulnerability} \times \text{Exposure}$) assumes that the three components interact symmetrically. In practice, extreme values of any single component may dominate outcomes in ways the multiplicative structure does not fully capture — for example, very high population density in a low-vulnerability ward may generate greater absolute health burden than a moderate-density, high-vulnerability ward, yet both could receive similar composite scores. The co-benefits analysis relies on the i-Tree Planting tool configured with US National Average parameters because Chennai is not available as a location in the tool. While emissions factors were localised (electricity: 710 kg CO₂ eq/MWh; fuel: 63 kg CO₂ eq/MMBtu), species growth curves and stormwater hydrology models embedded in i-Tree are calibrated to temperate North American conditions. The resulting physical co-benefit quantities should therefore be interpreted as order-of-magnitude estimates rather than precise projections for Chennai’s tropical climate. The equal weighting of the three 3-30-300 indicators assumes that tree visibility, canopy cover, and park access contribute independently and equivalently to neighbourhood livability. This is consistent with the original framework’s premise (Konijnendijk, 2021) but has not been empirically validated for Chennai’s specific urban context, where the relative importance of each indicator may differ.

9.3 Future Work

Several extensions would strengthen both the precision and the policy utility of this framework. On the hazard side, moving from single-date LST composites to multi-temporal analysis incorporating seasonal and diurnal variation would provide a more representative picture of heat exposure. Integrating humidity data — either from reanalysis products (ERA-5) or from ground-station records —

into a composite heat-stress index would address the most consistent methodological gap identified during expert consultation.

On the green coverage side, future iterations could enhance precision through fusion with higher-resolution commercial imagery, LiDAR-derived canopy height models, or citizen science tree inventories. The ongoing Tamil Nadu tree census, expected to produce results by 2027, would enable direct validation of the NDVI-based tree-visibility proxy against field-counted tree locations.

The co-benefits quantification would benefit from a Chennai-specific building energy model to estimate cooling-load reductions from canopy shading — a benefit category that the i-Tree tool could not estimate in its current configuration. A full Multi-Criteria Decision Analysis incorporating monetised and non-monetised co-benefits, structured around the ward-level budget categories recommended by expert consultees, would move the framework from identification of priority zones toward actionable investment appraisal.

Finally, for transferability to Gulf and South Asian coastal cities, adaptations for humidity weighting, cooling degree days, and mangrove-specific co-benefits would be required. Stress-testing the priority-zone recommendations across distinct hexagon typologies within Chennai — industrial, residential, coastal, peri-urban — would expose how local realities vary and strengthen the framework’s robustness for implementation.

Acknowledgements

We would like to express our deepest gratitude to everyone who supported, guided, and inspired us throughout this capstone project.

First and foremost, we are immensely grateful to our faculty advisor, **Professor David Lassarman**, for his invaluable mentorship, continuous guidance, and strategic direction. His support was instrumental in steering this project to completion.

We extend our sincere appreciation to our client, **Ammara Shariq** at the World Bank Group, for entrusting us with this incredible opportunity. Her vision, collaborative spirit, and unwavering support provided the foundation for our research.

We are also deeply thankful to **Professor Suzy Li**, who generously took the time to meet with us. Her insightful and professional advice significantly refined our analytical approach and dashboard building.

Our spatial framework builds upon the groundwork laid by **Zoe Baker, Nishan Sah, Gabe Hafemann, Sakina Mammadova, and Nandita Kannapadi**. We sincerely thank them for their pioneering work and their willingness to share their knowledge and tools.

We extend our deepest gratitude to the distinguished domain experts and stakeholders who generously shared their time and knowledge: **Christopher Goranson, Anshul Mishra, Rahul Agnihotri, Rajan Rawal, Intikhab Alam, Leigh Mante, Barjor Mehta, Suren Vakil, Rutul Joshi, Romit Sinha, and Ramayya Krishnan**. Your collective expertise, spanning Chennai's local context, complex datasets, ArcGIS methodologies, and financial frameworks, provided dedicated support when we needed it most. Your on-the-ground insights were vital in ensuring that our data-driven framework remains firmly grounded in real-world planning needs.

Finally, we would like to thank our audience, our peers, and everyone who attended our presentations and provided constructive feedback.

References

- Alves, A., Gersonius, B., Kapelan, Z., Vojinovic, Z., and Sanchez, A. (2019). Assessing the co-benefits of green-blue-grey infrastructure for sustainable urban flood risk management. *Journal of Cleaner Production*, 239:118102.
- Alves, A., Vojinovic, Z., Sanchez, A., and Martins, R. (2018). A carbon cost-benefit analysis framework for nature-based solutions in urban flood risk management. *Water Resources Management*, 32(8):2739–2759.
- Arunachalam, M., Saravanavel, J., and Kochuparampil, A. J. (2023). PCA-based approach for mapping social vulnerability to hazards in the Chennai metropolitan area, east coast of India. *Annals of GIS*, 29(4):529–552.
- Ballester, J., Quijal-Zamorano, M., Turrubiates, M., and Achebak, H. (2025). Sustainable urban mitigation strategies regarding heat-related mortalities. *Theoretical and Applied Climatology*.
- Biasin, A., Datola, G., and De Angelis, E. (2023). NBS modeling and cost-benefit analysis for urban heat island and pluvial flood risk reduction: The case of Turin. *Land*, 12(2):280.
- Bowler, D. E., Buyung-Ali, L., Knight, T. M., and Pullin, A. S. (2010). Urban greening to cool towns and cities: A systematic review of the empirical evidence. *Landscape and Urban Planning*, 97(3):147–155.
- Carlson, T. N. and Ripley, D. A. (1997). On the relation between NDVI, fractional vegetation cover, and leaf area index. *Remote Sensing of Environment*, 62(3):241–252.
- Carlyle-Moses, D. E. and Gash, J. H. C. (2011). Rainfall interception loss by forest canopies. pages 407–423.
- Center for Watershed Protection (2016). Urban stormwater forestry.
- Chelli, S., Brander, L., and Geneletti, D. (2024). Cost-benefit analysis of urban nature-based solutions: A systematic review. *Ecosystem Services*, 66:101608.
- Chen, L., Doshi, V., Masood, F., and Zhou, Y. (2026a). Chennai co-benefits analysis matrix. Unpublished internal document, Carnegie Mellon University / World Bank Group.
- Chen, L., Doshi, V., Masood, F., and Zhou, Y. (2026b). SUHII analysis for Chennai. ArcGIS StoryMap, Carnegie Mellon University / World Bank Group.
- Cutter, S. L., Boruff, B. J., and Shirley, W. L. (2003). Social vulnerability to environmental hazards. *Social Science Quarterly*, 84(2):242–261.
- de Bont, J., Nori-Sarma, A., Stafoggia, M., Banerjee, T., Ingole, V., Jaganathan, S., Mandal, S., Rajiva, A., Krishna, B., Kloog, I., Lane, K., Mall, R. K., Tiwari, A., Wei, Y., Wellenius, G. A., Prabhakaran, D., Schwartz, J., Prabhakaran, P., and Ljungman, P. (2024). Impact of heatwaves on all-cause mortality in India: A comprehensive multi-city study. *Environment International*, 184:108461.
- Doan, M. K., Hill, R., Hallegatte, S., Corral, P., Brunckhorst, B., Nguyen, M., Freije-Rodriguez, S., and Naikal, E. (2023). Counting people exposed to, vulnerable to, or at high risk from climate shocks: A methodology. Policy Research Working Paper 10619, World Bank Group.
- Gonzalez-Casares, M., Endreny, T. A., et al. (2025). From site-scale to landscape-scale: Stormwater benefits of urban trees with SWMM-Canopy. *Urban Water Journal*.

- Hilbert, D. R., Roman, L. A., Koeser, A. K., Vogt, J., and van Doorn, N. S. (2019). Urban tree mortality: A literature review. *Arboriculture & Urban Forestry*, 45(5):167–200.
- Institute of Economic Growth (2023). Reassessment of national parameters for project appraisal in India. Technical report, NITI Aayog.
- IPCC (2014). Climate change 2014: Impacts, adaptation, and vulnerability. Part A: Global and sectoral aspects. In *Contribution of Working Group II to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press.
- IPCC (2022). Climate change 2021: Impacts, adaptation, and vulnerability. In *Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Cambridge University Press.
- Konijnendijk, C. C. (2021). Promoting health and wellbeing through urban forests: Introducing the 3–30–300 rule. IUCN Urban Alliance.
- Konijnendijk, C. C. (2023). Evidence-based guidelines for greener, healthier, more resilient neighbourhoods: Introducing the 3–30–300 rule. *Journal of Forestry Research*, 34:1399–1411.
- Kuehler, E., Hathaway, J., and Tirpak, A. (2017). Quantifying the benefits of urban forest systems as a component of the green infrastructure stormwater treatment network. *Ecohydrology*, 10(3):e1813.
- Li, Y. et al. (2025). Tree canopy halves the urban heat island effect globally. *ResearchGate*.
- Middel, A., Häb, K., Brazel, A. J., Martin, C. A., and Guhathakurta, S. (2014). Impact of urban form and design on mid-afternoon microclimate in Phoenix local climate zones. *Landscape and Urban Planning*, 122:16–28.
- Muthiah, K., Mathivanan, M., and Duraisekaran, E. (2022). Dynamics of urban sprawl on the peri-urban landscape and its relationship with urban heat island in Chennai metropolitan area, India. *Arabian Journal of Geosciences*, 15:1694.
- Nithila Devi, N., Sridharan, B., and Kuiry, S. N. (2019). Impact of urban sprawl on future flooding in Chennai city, India. *Journal of Hydrology*, 574:486–496.
- Nowak, D. J., Crane, D. E., Stevens, J. C., Hoehn, R. E., Walton, J. T., and Bond, J. (2014). Tree and forest effects on air quality and human health in the United States. *Environmental Pollution*, 193:119–129.
- Nowak, D. J., Hirabayashi, S., Bodine, A., and Hoehn, R. (2013). Modeled PM_{2.5} removal by trees in ten U.S. cities and associated health effects. *Environmental Pollution*, 178:395–402.
- Office of the Registrar General & Census Commissioner, India (2011). Census of India 2011. Ministry of Home Affairs, Government of India.
- OpenStreetMap contributors (2024). OpenStreetMap.
- Pope, C. A., Burnett, R. T., Thun, M. J., Calle, E. E., Krewski, D., Ito, K., and Thurston, G. D. (2002). Lung cancer, cardiopulmonary mortality, and long-term exposure to fine particulate air pollution. *JAMA*, 287(9):1132–1141.
- Raj, S. and Yun, G. Y. (2024). Influence of selection of rural reference area for quantifying the surface urban heat islands intensity in major South Korean cities. *Architectural Science Review*, 67(4):345–356.

- Ruangpan, L., Vojinovic, Z., Di Sabatino, S., Leo, L. S., Capobianco, V., Aronica, G. T., and Sanchez, A. (2020). Nature-based solutions for hydro-meteorological risk reduction: A state-of-the-art review. *Natural Hazards and Earth System Sciences*, 20:243–270.
- Ruangpan, L., Vojinovic, Z., Di Sabatino, S., and Sanchez, A. (2024). Co-benefit-aware life-cycle cost-benefit analysis of nature-based solutions at river-basin scale. *Journal of Environmental Management*, 349:119499.
- Sah, N. (2025). SUHII analysis tool. GitHub.
- Selbig, W. R., Loheide, S. P., Shuster, W., Scharenbroch, B. C., Coville, R. C., Kruegler, J., Avery, W., Haefner, R., and Nowak, D. (2022). Quantifying the stormwater runoff reduction benefits of urban street trees: A case study. Fact Sheet 2022–3074, U.S. Geological Survey.
- Selmi, W., Weber, C., Rivière, E., Blond, N., Mehdi, L., and Nowak, D. (2016). Air pollution removal by trees in public green spaces in Strasbourg city, France. *Urban Forestry & Urban Greening*, 17:192–201.
- Stewart, I. D. and Mills, G. (2021). *The Urban Heat Island: A Guidebook*. Elsevier.
- TNGCC and ICLEI South Asia (2026). Urban heat island assessment and strategic guidelines for urban cooling in Tamil Nadu. Technical report, Tamil Nadu Green Climate Company, Department of Environment, Climate Change and Forests, Government of Tamil Nadu.
- U.S. Geological Survey (2024). Landsat 9. USGS Landsat Missions.
- USDA Forest Service (2024). i-Tree planting tool. USDA Forest Service.
- Weier, J. and Herring, D. (2000). Measuring vegetation (NDVI & EVI). NASA Earth Observatory.
- Xiao, Q. and McPherson, E. G. (1998). Rainfall interception by Santa Monica’s municipal urban forest. *Urban Ecosystems*, 2(4):291–302.
- Xiao, Q. and McPherson, E. G. (2002). Rainfall interception by Sacramento’s urban forest. *Journal of Arboriculture*, 28(4):235–244.
- Zabret, K. and Šraj, M. (2023). Rainfall interception by urban trees: Canopy intensity attenuation in Ljubljana. *Urban Forestry & Urban Greening*.
- Zhou, D., Xiao, J., Bonafoni, S., Berger, C., Deilami, K., Zhou, Y., Frolking, S., Yao, R., Qiao, Z., and Sobrino, J. A. (2019). Satellite remote sensing of surface urban heat islands: Progress, challenges, and perspectives. *Remote Sensing*, 11(1):48.
- Zhou, D., Zhao, S., Liu, S., Zhang, L., and Zhu, C. (2014). Surface urban heat island in China’s 32 major cities: Spatial patterns and drivers. *Remote Sensing of Environment*, 152:51–61.
- Ziter, C. D. et al. (2025). Increasing tree canopy lowers urban air temperature by up to 1.5 °C. *npj Urban Sustainability*.
- Ziter, C. D., Pedersen, E. J., Kucharik, C. J., and Turner, M. G. (2019). Scale-dependent interactions between tree canopy cover and impervious surfaces reduce daytime urban heat during summer. *Proceedings of the National Academy of Sciences*, 116(15):7575–7580.
- Publicly available artificial intelligence tools, if any, were used responsibly in the scoping, problem-solving and reporting of this project.*